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(54) **INSECT-DERIVED ASPARTATE
DECARBOXYLASES AND VARIANTS
THEREOF FOR IMPROVED BETA-ALANINE
PRODUCTION**

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(57) **ABSTRACT**

Provided are N-terminally truncated variants of insect aspartate 1-decarboxylases that exhibit improved performance for beta-alanine production.

Specification includes a Sequence Listing.

A circular phylogenetic tree showing the relationships between various insect orders. The tree is divided into several major groups, each labeled with a large, bold, black text label. The groups are: Butterflies, Moths; Wasps, Bees, Ants; Beetles; Fleas, Roaches, Termites; Bugs, Aphids; Mayflies, Thrips, Lice; Flies; and Mosquitoes. Each group contains a list of species names, with some species names highlighted in a light blue box. The species names are arranged in a circular pattern around the tree, with some species names appearing multiple times. The tree is rooted at the top, with the root node labeled 'ADC'. The tree is a cladogram, showing the evolutionary relationships between the species. The branches are labeled with the names of the species, and the nodes are labeled with the names of the species. The tree is a circular phylogenetic tree, showing the relationships between various insect orders. The tree is rooted at the top, with the root node labeled 'ADC'. The tree is a cladogram, showing the evolutionary relationships between the species. The branches are labeled with the names of the species, and the nodes are labeled with the names of the species. The tree is a circular phylogenetic tree, showing the relationships between various insect orders. The tree is rooted at the top, with the root node labeled 'ADC'. The tree is a cladogram, showing the evolutionary relationships between the species. The branches are labeled with the names of the species, and the nodes are labeled with the names of the species.

Fig. 2

Aa2ADC (10)	MPANGMFQVALQVIDDS---NVSSGSDSAGVSEDEDVQLFCSMGNTIAP---KPLKKSITRT	56
Aa3ADC (11)	MPANGMFQVALQVIDDS---NVSSGSDSAGVSEDEDVQLFCSKGNTIVT---KPLKKSISKI	56
CtADC (2)	MPANGMGLDVALQVIEDA---NLSSGSDSAGVSEDEDVQLFSTTGNIVSS---KPLKPKALKP	56
CqADC (9)	MPANGMFQVALQVIEDA---NLSSGSDSAGVSEDEDVQLFCTTGNVVS---KPLKRPALKP	56
Aa4ADC (13)	MPATGVSSIGLEVQEE-PATYASGSDSAGVSEDEDVQLFVNGAHGLTSVAFVRKP-EM	58
AdADC (14)	MPATGVSSIGLEVHEE-PATYASGSDSAGVSEDEDVQLFVNGAHGVTRVAPAARKA-EM	58
AsADC (4)	MPANGVCSVGLEVIEDN-ATYASGSDSAGVSEDEDVQLFVSGADRVTSVLPKKSIR---	57
AsADC (12)	MPANGVMSVELEVIEDVATTYASGSDSAGVSEDEDVQLFVSGARHISVVPPLKRAV-ET	59
As2ADC (15)	MPANGVCSVGLEVIEDNAATYASGSDSAGVSEDEDVQLFVNGADRVTSSVSLPKKSTEA	60

Aa2ADC (10)	K-----DEEFSKTAKANEKRYASLPNREQHQQLTDFLSEVLNNAVFNATERANKV	107
Aa3ADC (11)	K-----DEEFSKTAKANEKRYASLPNREQHQQLTDFLSEVLNNAVFNATERANKV	107
CtADC (2)	AT-----KDEDQNKTKANEKRYASLPNREQHQQLTDFLSEVLNCAIFNATORSNKV	108
CqADC (9)	VT-----TVKDEDQNKTKANEKRYASLPNREQHQQLTDFLSEVLNNAVFNATERANKV	110
Aa4ADC (13)	RGKLS-----LDESAAIDPKRYASLPNREQHQQLTEFLTEVLNSAVFNATERANKV	110
AdADC (14)	RGKLS-----LDESAAIDPKRYASLPNREQHQQLTEFLTEVLNSAVFNATERANKV	110
AsADC (4)	---KASQVDEQAAAAAAAVSEKRYASLPNREQHQQLTQFLTEVLNSAVFNATERANKV	115
AsADC (12)	RGKGTQLQSPASEGAAAAEVSEKRYASLPNREQHQQLTDFLSEVLNSAVFNATERANKV	119
As2ADC (15)	RGKLSQH-----GDDGKPAVAEKRYASLPNREQHQQLTEFLTEVLNSAVFNATERANKV	115

Aa2ADC (10)	LNWVDPEQLKRTLDLLELKDEPDSEHEKLELLELTRATIRHSVKTGHPYFMNQLFSSVDPYGFA	167
Aa3ADC (11)	LNWVDPEQLKRTLDLLELKDEPDSEHEKLELLELTRATIRHSVKTGHPYFMNQLFSSVDPYGFA	167
CtADC (2)	LNWVDPEELKRSIDLLELKDEPDSEHEKLELLELTRATIRHSVKTGHPYFMNQLFSSVDPYGFA	168
CqADC (9)	LNWVDPEELKRSIDLLELKDEPDSEHEKLELLELTRATIRHSVKTGHPYFMNQLFSSVDPYGFA	170
Aa4ADC (13)	LNWVDPEELSRITLDLLELKDEPDSEHEKLELLELTRATIRHSVKTGHPYFMNQLFSSVDPYGFA	170
AdADC (14)	LNWVDPEELSRITLDLLELKDEPDSEHEKLELLELTRATIRHSVKTGHPYFMNQLFSSVDPYGFA	170
AaADC (4)	LNWVDPEELQRTLDLLELKDEPDSEHEKLELLELTRATIRHSVKTGHPYFMNQLFSSVDPYGFA	175
AsADC (12)	LNWVDPEELKRTLDLLELKDEPDSEHEKLELLELTRATIRHSVKTGHPYFMNQLFSSVDPYGFA	179
As2ADC (15)	LNWVDPEELKRTLDLLELKDEPDSEHEKLELLELTRATIRHSVKTGHPYFMNQLFSSVDPYGFA	175

Aa2ADC (10)	GQILTDAINPVSVYTFEVSPVFLMEEVVLKEMRTIVGYPDGAGDGIFCPGGSMANGYSIS	227
Aa3ADC (11)	GQILTDAINPVSVYTFEVSPVFLMEEVVLKEMRTIVGYPDGAGDGIFCPGGSMANGYSIS	227
CtADC (2)	GQVLTDAINPVSVYTFEVSPVFLMEEVVLKEMRTIVGYPDGAGDGIFCPGGSMANGYSIS	228
CqADC (9)	GQCLTDAINPVSVYTFEVSPVFLMEEVVLKEMRTIVGYPDGAGDGIFCPGGSMANGYSIS	230
Aa4ADC (13)	GQVLTDAINPVSVYTFEVSPVFLMEEVVLKEMRTIVGYPDGAGDGIFCPGGSMANGYSIS	230
AdADC (14)	GQVLTDAINPVSVYTFEVSPVFLMEEVVLKEMRTIVGYPDGAGDGIFCPGGSMANGYSIS	230
AaADC (4)	GQVLTDAINPVSVYTFEVSPVFLMEEVVLKEMRTIVGYPDGAGDGIFCPGGSMANGYSIS	235
AsADC (12)	GQVLTDAINPVSVYTFEVSPVFLMEEVVLKEMRTIVGYPDGAGDGIFCPGGSMANGYSIS	239
As2ADC (15)	GQVLTDAINPVSVYTFEVSPVFLMEEVVLKEMRTIVGYPDGAGDGIFCPGGSMANGYSIS	235

Aa2ADC (10)	CARFKHMPDVKT-----	239
As3ADC (11)	CARFKHMPDVKT-----	239
CtADC (2)	CARFKHMPDVKT-----	240
CqADC (9)	CARFKHMPDVKT-----	242
Aa4ADC (13)	CARFKHMPDIKA-----	242
AdADC (14)	CARFKHMPDIKESFRTMHLITLIESAGYGMTIVSQHVTTVVAIAIKVHRQRSTGCYTR	290
AaADC (4)	CARFKHMPDIKT-----	247
AsADC (12)	CARYKHMPDVKA-----	251
As2ADC (15)	CARFKHMPDIKT-----	247

Aa2ADC (10)	-----KGLHSLPRVIFTSEDAHYSVKKLASFMSGIGSDNVYPIRTDAIGKIR	286
Aa3ADC (11)	-----KGLHSLPRVIFTSEDAHYSVKKLASFMSGIGSDNVYPIRTDAIGKIR	286
CtADC (2)	-----KGLHSLPRVIFTSEDAHYSIKKLASFMSGIGSDNVYPIRTDAVGRIQ	287
CqADC (9)	-----KGLHSLPRVIFTSEDAHYSIKKLASFMSGIGSDNVYPIRTDAVGRIQ	289
Aa4ADC (13)	-----KGLHALPRVIFTSEDAHYSIKKLASFMSGIGSDNVYPIKTDEIGKIC	289
AdADC (14)	SWLIETIGNQASAKGLHALPRVIFTSEDAHYSIKKLASFMSGIGSDNVYPIKTDEIGKIR	350
AaADC (4)	-----KGLHALPRVIFTSEDAHYSVKKLASFMSGIGSDNVYAIKTDNVGRI	294
AsADC (12)	-----KGLHALPRVIFTSEDAHYSVKKLASFMSGIGSDNVYAIKTDNVGRI	298
As2ADC (15)	-----KGLHALPRVIFTSEDAHYSVKKLASFMSGIGSDNVYAIKTDNVGRI	294

Fig. 2 (cont'd)

Aa2ADC (10)	VDHLESEILRAKAEAGVPMVVSATAGTTVIGAFDPLEQIADLCCKYNLWMMHVDAAWGGGA	346
Aa3ADC (11)	VDHLESEILRAKSEGAVPFMVSATAGTTVIGAFDPLEQIADLCCKYNLWMMHVDAAWGGGA	346
CtADC (2)	PDHLEAEILRAKSEGAVPFMVSATAGTTVIGAFDPLEQIADLCCKYNLWMMHVDAAWGGGA	347
CqADC (9)	PDHLEAEILRAKSEGALPFMVSATAGTTVIGAFDPLEQIADLCCKYNLWMMHVDAAWGGGA	349
Aa4ADC (13)	VDHLESEILRAKAEAGSPFMVVSATAGTTVIGAFDPLEQIADLCCKYQLWPFHVDAAWGGGA	349
AdADC (14)	VDHLESEILRAKAEAGALPFMVSATAGTTVIGAFDPLEQIADLCCKYQLWPFHVDAAWGGGA	410
AaADC (4)	VEHLESEILRAKSEGALPFMVSATAGTTVIGAFDPLEQIADLCCKYNLWMMHVDAAWGGGA	354
AsADC (12)	VDHLESEILRAKQEGALPFMVSATAGTTVIGAFDPLEQIADLCCKYNLWMMHVDAAWGGGA	358
As2ADC (15)	IEHLESEILRAKAEAGALPFMVSATAGTTVIGAFDPLEQIADLCCKHNLWMMHVDAAWGGGA	354

Aa2ADC (10)	LMSKKYRSLKGIERSDSVTWNPBKLLAAPQQCSTFLTRHEGILSECHSTNATYLFQKDK	406
Aa3ADC (11)	LMSKKYRSLKGIERSDSVTWNPBKLLAAPQQCSTFLTRHEGILSECHSTNATYLFQKDK	406
CtADC (2)	LMSKKYRSLKGIERSDSVTWNPBKLLAAPQQCSTFLTRHEGILSGCHSTNATYLFQKDK	407
CqADC (9)	LMSKKYRSLKGIERSDSVTWNPBKLLAAPQQCSTFLTRHEGILSGCHSTNATYLFQKDK	409
Aa4ADC (13)	LMSKKYRSLKGIERSDSVTWNPBKLLAAPQQCSTLLTRHPNLLSECHSTNATYLFQKDK	409
AdADC (14)	LMSKKYRSLKGIERSDSVTWNPBKLLAAPQQCSTLLTRHPNLLSECHSTNATYLFQKDK	470
AaADC (4)	LMSKKYRSLKGIERSDSVTWNPBKLLAAPQQCSTLLTRHNLAEASTNATYLFQKDK	414
AsADC (12)	LMSKKYRSLKGIERSDSVTWNPBKLLAAPQQCSTLLTRHNLSECHSTNATYLFQKDK	418
As2ADC (15)	LMSKKYRSLKGIERSDSVTWNPBKLLAAPQQCSTLLTRHNLSECHSTNATYLFQKDK	414

Aa2ADC (10)	FYDTQYDTGDKHIQCGRRADVLFKFWFMMWRAGTSGLEQHIDKVFENAEHFTNSIKARPGF	466
Aa3ADC (11)	FYDTQYDTGDKHIQCGRRADVLFKFWFMMWRAGTSGLEQHIDKVFENAEHFTSSIKARPGF	466
CtADC (2)	FYDTQYDTGDKHIQCGRRADVLFKFWFMMWRAGTSGLEQHIDKVFETAEYFTNSIKARPGF	467
CqADC (9)	FYDTQYDTGDKHIQCGRRADVLFKFWFMMWRAGTSGFEGHIDKVFENAEYFTNSIKARPGF	469
Aa4ADC (13)	FYDTQYDTGDKHIQCGRRADVLFKFWFMMWRAGTSGFEGHIDKVFENAEYFTNSIKARPGF	469
AdADC (14)	FYDTQYDTGDKHIQCGRRADVLFKFWFMMWRAGTSGFEGHIDKVFENAEYFTNSIKARPGF	530
AaADC (4)	FYDTRYDTGDKHIQCGRRADVLFKFWFMMWRAGTAGFEAHIDKVFENAEHFTSSIKARPGF	474
AsADC (12)	FYDTRYDTGDKHIQCGRRADVLFKFWFMMWRAGTAGFEAHIDKVFENAEHFTSSIKARPGF	478
As2ADC (15)	FYDTQYDTGDKHIQCGRRADVLFKFWFMMWRAGTAGFEAHIDKVFENAEHFTNSIKARPGF	474

Aa2ADC (10)	EMVVENPECTNVCFWYVPPGLRNVPRDSAEFTEERLHKVAPKVKERMREGSMMITYQPIH	526
Aa3ADC (11)	EMVVENPECTNVCFWYVPPGLRNVPRDSAEFTEERLHKVAPKVKERMREGSMMITYQPIH	526
CtADC (2)	EMVVENPECTNVCFWYVPPGLRQVPRDSAEFTEERLHKVAPKVKERMREGSMMITYQPIH	527
CqADC (9)	EMVVENPECTNVCFWYVPPGLRQVPRDSAEFTEERLHKVAPKVKERMREGSMMITYQPIH	529
Aa4ADC (13)	EMVVENPECTNVCFWYVPPGLRDMARDSAEYTERLHKVAPKVKERMREGSMMITYQPIH	529
AdADC (14)	EMVVENPECTNVCFWYVPPGLRDMARDSAEYTERLHKVAPKVKERMREGSMMITYQPIH	590
AaADC (4)	EMVVENPECTNVCFWYVPPGLRGVPRDSAEYTERLHKVAPKVKERMREGSMMITYQPIH	534
AsADC (12)	EMVVENPECTNVCFWYVPPGLRNVPRDSAEFTEERLHKVAPKVKERMREGSMMITYQPIH	538
As2ADC (15)	EMVVENPECTNVCFWYVPPGLRNVPRDSAEFTEERLHKVAPKVKERMREGSMMITYQPIH	534

Aa2ADC (10)	DKPNFFRLVLQNSALDKSDMNYIIDEIERLAADL---	560
Aa3ADC (11)	DKPNFFRLVLQNSALDKSDMNYIIDEIERLAADIKP---	562
CtADC (2)	DKPNFFRLVLQNSALDKSDMNYIIDEIERLASDL---	561
CqADC (9)	DKPNFFRLVLQNSALDKSDMNYIIDEIERLASDL---	563
Aa4ADC (13)	DKPNFFRLVLQNSALDKSDMNYIIDEIERLGKDL---	563
AdADC (14)	DKPNFFRLVLQNSALDKSDMNYIIDEIERLGKDL---	624
AaADC (4)	DKPNFFRLVLQNSALDKSDMNYIIDEIERLGKDL---	568
AsADC (12)	DKPNFFRLVLQNSALDKSDMNYIIDEIERLGKDL---	572
As2ADC (15)	DKPNFFRLVLQNSALDKSDMNYIIDEIERLGKDL---	568

CtADC	(2)	MFTNGMLDVALQVIEDA---NLSSGSDSAGVSEDEDVQLFSTTCGNIVSS---KPLKKPALKP	56
AaADC	(4)	MFANGVCSVGLEVIEDN-ATYASGSDSAGVSEDEDVQLFPVSGADRVTISVLPRKSDIR---	57
AtADC	(3)	MFANGQLEDG-EHLIDEPATYSDAVA-----SSSDDETVOQSNDRSI-----	42
TcADC	(1)	MFATGEDQDLVQDLIEEPATFSDAVL-----SSDEELFHQCKPKPAP-----	42
Aa2ADC	(10)	MPANGMEDVALQVIDDS---NVSSGSDSAGVSEDEDVQLFCSMGNTIAP---KELKKSITKT	56
Aa3ADC	(11)	MFANGMFDVALQVIDDS---NVSSGSDSAGVSEDEDVQLFCSKGNTIVP---KFLKKSISKI	56
CqADC	(9)	MFINGMFDVALQVIEDA---NLSSGSDSAGVSEDEDVQLFCTTGNVVSS---KELKKPSLKF	56
Aa4ADC	(13)	MFATGVSSIGLEVQEE-PATYASGSDSAGVSEDEDVQLFVNAGHLISVAFAVRKE-EM	58
AdADC	(14)	MFATGVSSIGLEVHEE-PATYASGSDSAGVSEDEDVQLFVNGAHGVTRVPAARKA-EM	58
AsADC	(12)	MPANGVNSVELEVIEDVAITYASGSDSAGVSEDEDVQLFVSGAHHSVPELKAV-ET	59
As2ADC	(15)	MPANGVCSVGLEVIEDNAATYASGSDSAGVSEDEDVQLFVNGADRVTSVSSLPKKSTE	60
TmADC	(6)	MPARGEQDDVVQDIITEEPATYGAIL---SSDDEVYTTFKERP-----	40
AvADC	(5)	MFATGEQDDLVDQDIITEEPATYSDAVL---SSDDEVCVRYSSQSDTN-----	43
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CtADC	(2)	AT-----KDEDQNKTKANAKRYASLPNREHQRFLLIDFLSEVINCAIFNATDRSNKV	108
AaADC	(4)	--RASQVDEQAATAAAAVSEKRYASLPNREHQRFLLTQFLTEVLNSAVFNATDRANKV	115
AtADC	(3)	-----RMKRTATTATGNLATFESLPSKAHHEEFIRSCMDVILKEAVFDGTNRNNPV	92
TcADC	(1)	-----TYSP-VSRHFESLENREHQRFLLTQFLTEVLNSAVFNATERANKV	88
Aa2ADC	(10)	K-----DEEFSKTAKANERKYASLPNREHQRFLLTFLSEVLNNAVFNATERANKV	107
Aa3ADC	(11)	K-----DEEFSKTAKANERKYASLPNREHQRFLLTFLSEVLNNAVFNATERANKV	107
CqADC	(9)	VT-----TVRDEDQNKMKTNAKRYASLPNREHQRFLLIDFLSEVLNNALFNATDRSNKV	110
Aa4ADC	(13)	RGKLS-----LDESAADDRKRYASLPNREHQRFLLTEFLTEVLNSAVFNATDRANKV	110
AdADC	(14)	RGKLS-----LDESAADDRKRYASLPNREHQRFLLTEFLTEVLNSAVFNATDRANKV	110
AsADC	(12)	RKGKTQLQGPFASEGAATAAEVSEKRYASLPNREHQRFLLIDFLTEVLNSAVFNATDRANKV	119
As2ADC	(15)	RGKLSQLH----GDDCKPFAAEKRYASLPNREHQRFLLTEFLTEVLNSAVFNATDRSNKV	115
TmADC	(6)	-----LTQFYQP-SQRASFESLPNREHQRFLLIKRAVEILLKDAVEGTGRNNRV	89
AvADC	(5)	-----NSSFYQTATKRLASFESLPNREHQRFLLIKRCALILIREAVEFGTNRKNRV	93
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CtADC	(2)	LNWVDPEELKRSLDLSLNDEPOSDEKLLLELARATIDRSVRTGHYPFMNQLFSSVDPYGFA	168
AaADC	(4)	LNWVDPEELQRTLIDLAIKDPEPTHEKLLLELTRATRHSVKTGHPYFMNQLFSSVDPYGFA	175
AtADC	(3)	LNFVNPEELQSKVNFKLKTAFTHEDLLTKLTDTRIYSVKTGHPYFMNQLFSSLDPYGLV	152
TcADC	(1)	LQWREPEELRLRLDMDFGVSAFTHEELLLEVLLKVVTVSVKTHGHPYFMNQLFSAVDYGLV	148
Aa2ADC	(10)	LNWVDPEELKRTLIDLELKDEFDSHEKLLLELTRATIKHSVTKGHPYFMNQLFSSVDPYGFA	167
Aa3ADC	(11)	LNWVDPEELKRTLIDLELKDEFDSHEKLLLELTRATIKHSVTKGHPYFMNQLFSSVDPYGFA	167
CqADC	(9)	LNWVDPEELKRSLDLSLKAEPDSDEKLLLELARATIDHSVRTGHYPFMNQLFSSVDVYGFA	170
Aa4ADC	(13)	LNWVDPEELSRTLIDLAIKDEFDIERLLELTRATRHSVRTGHYPFMNQLFSSVDPYGFA	170
AdADC	(14)	LNWVDPEELSRTLIDLAIKDEFDIERLLELTRATRHSVRTGHYPFMNQLFSSVDPYGFA	170
AsADC	(12)	LNWVDPEELKRTLIDLAIKQEPDTHEKLLLELTRATRHSVRTGHYPFMNQLFSSVDPYGFA	179
As2ADC	(15)	LNWVDPEELKRTLIDLAIKDEPDOTHEKLLLELTRATRHSVRTGHYPFMNQLFSSVDPYGFA	175
TmADC	(6)	LQWTCPHEELSRLEMEFGLKNGPSTHEELBLIKKVVNYSVRTGHYPFMNQLFSSLDPYGLV	149
AvADC	(5)	LQWNSSPEELQKLMDFTLRTSPSSHDELDDLRLNTVMNYSVRTGHYPFMNQLFSSLDPYGLV	153
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**INSECT-DERIVED ASPARTATE
DECARBOXYLASES AND VARIANTS
THEREOF FOR IMPROVED BETA-ALANINE
PRODUCTION**

**CROSS-REFERENCE TO RELATED
APPLICATIONS**

[0001] This application is the U.S. National Stage of International Application No. PCT/CN2022/079042, filed on Mar. 3, 2022, which designates the U.S., published in English, and claims the priority benefit of International Application No. PCT/CN2021/078949, filed on Mar. 3, 2021.

[0002] The present description relates to biological processes for the production of beta-alanine. More particularly, described herein are insect aspartate 1-decarboxylase (ADC) enzymes and variants thereof that are particularly advantageous for beta-alanine production from L-aspartate.

BACKGROUND

[0003] Beta-alanine, also known as beta-aminopropionic acid or 3-aminopropionic acid, is a naturally occurring amino acid in which the amino group is at the beta-position from the carboxylate group. Beta-alanine is a multi-purpose organic synthetic raw material, mainly used to synthesize pantothenic acid and calcium pantothenate, carnosine, pamidronate, balsalazide, etc. It is widely used in medicine, feed, food and other fields, and has a large market demand. At industrial scales, beta-alanine is currently produced by chemical processes involving harsh reaction conditions with safety concerns, high equipment costs, and environmental pollution. The production of beta-alanine by safer and more environmentally friendly biological processes have been greatly hampered by enzymes having poor activity, expression, and/or stability, thereby making such processes not commercially viable in comparison to chemical synthesis approaches. Therefore, improved enzymes useful for the biological production of beta-alanine are highly desirable.

SUMMARY

[0004] In one aspect, described herein is a recombinant truncated insect aspartate 1-decarboxylase (ADC), the truncated insect ADC lacking a sufficient number of contiguous residues within the amino terminal region of a corresponding full length wild-type insect ADC such that the truncated ADC exhibits increased conversion of aspartate to beta-alanine as compared to the corresponding full length wild-type insect ADC.

[0005] In a further aspect, described herein is a recombinant protein having aspartate 1-decarboxylase activity, the recombinant protein comprising an amino acid sequence at least 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, or 99% identical overall to:

[0006] (a) position 72 to 561 of the amino acid sequence of CtADC set forth in SEQ ID NO: 2;

[0007] (b) position 79 to 568 of the amino acid sequence of AaADC set forth in SEQ ID NO: 4;

[0008] (c) position 56 to 544 of the amino acid sequence of AtADC set forth in SEQ ID NO: 3;

[0009] (d) position 52 to 540 of the amino acid sequence of TcADC set forth in SEQ ID NO: 1;

[0010] (e) position 71 to 560 of the amino acid sequence of Aa2ADC set forth in SEQ ID NO: 10;

[0011] (f) position 71 to 562 of the amino acid sequence of Aa3ADC set forth in SEQ ID NO: 11;

[0012] (g) position 74 to 563 of the amino acid sequence of CqADC set forth in SEQ ID NO: 9;

[0013] (h) position 72 to 561 of the amino acid sequence of Aa4ADC set forth in SEQ ID NO: 13;

[0014] (i) position 74 to 624 of the amino acid sequence of AdADC set forth in SEQ ID NO: 14;

[0015] (j) position 83 to 572 of the amino acid sequence of AsADC set forth in SEQ ID NO: 12;

[0016] (k) position 72 to 561 of the amino acid sequence of As2ADC set forth in SEQ ID NO: 15;

[0017] (l) position 53 to 541 of the amino acid sequence of TmADC set forth in SEQ ID NO: 6;

[0018] (m) position 57 to 572 of the amino acid sequence of AvADC set forth in SEQ ID NO: 5; or

[0019] (n) position 1 to 491 of the amino acid sequence of SEQ ID NO: 29.

[0020] In a further aspect, described herein is a polynucleotide comprising a nucleic acid sequence encoding the recombinant truncated insect ADC described herein, or the recombinant protein described herein.

[0021] In a further aspect, described herein is an expression cassette comprising the isolated or recombinant polynucleotide described herein operably linked to a promoter that is heterologous with respect to the insect ADC.

[0022] In a further aspect, described herein is a host cell that expresses the recombinant truncated insect ADC described herein, the recombinant protein described herein, and/or is transformed with or engineered to comprise the polynucleotide described herein or the expression cassette described herein.

[0023] In a further aspect, described herein is a process for the production of beta-alanine, the process comprising: (a) providing an ADC enzyme source which is the truncated insect ADC described herein, the recombinant protein described herein, and/or the host cell described herein; (b) contacting the ADC enzyme source with a source of aspartate under conditions enabling the ADC enzyme source to catalyze the conversion of the aspartate to beta-alanine; and (c) isolating and/or concentrating the beta-alanine produced.

[0024] In a further aspect, described herein is a composition comprising beta-alanine produced by the process described herein.

General Definitions

[0025] Headings, and other identifiers, e.g., (a), (b), (i), (ii), etc., are presented merely for ease of reading the specification and claims. The use of headings or other identifiers in the specification or claims does not necessarily require the steps or elements be performed in alphabetical or numerical order or the order in which they are presented.

[0026] The use of the word “a” or “an” when used in conjunction with the term “comprising” in the claims and/or the specification may mean “one” but it is also consistent with the meaning of “one or more”, “at least one”, and “one or more than one”.

[0027] The term “about” is used to indicate that a value includes the standard deviation of error for the device or method being employed in order to determine the value. In general, the terminology “about” is meant to designate a possible variation of up to 10%. Therefore, a variation of 1, 2, 3, 4, 5, 6, 7, 8, 9 and 10% of a value is included in the term

“about”. Unless indicated otherwise, use of the term “about” before a range applies to both ends of the range.

[0028] As used herein, the terms “comprising” (and any form of comprising, such as “comprise” and “comprises”), “having” (and any form of having, such as “have” and “has”), “including” (and any form of including, such as “includes” and “include”) or “containing” (and any form of containing, such as “contains” and “contain”) are inclusive or open-ended and do not exclude additional, unrecited elements or process/method steps.

[0029] As used herein, the term “beta-alanine” includes beta-alanine as well as beta-alanine salts (e.g., calcium, sodium, or potassium beta-alanine salt).

BRIEF DESCRIPTION OF THE DRAWINGS

[0030] In the appended drawings:

[0031] FIG. 1 shows a phylogenetic tree of ADC enzymes from different insect species grouped by 85% sequence identity. Activity data for some ADCs tested in Example 2 are shown.

[0032] FIG. 2 shows an alignment of amino acid sequences of ADCs identified from nine different mosquito species. The broken line delineates a region in the N terminal portion that is poorly conserved across mosquito ADCs. A glycine residue at position 96 that is unique to CtADC is highlighted in black. SEQ IN NOS corresponding to each sequence are in parentheses.

[0033] FIG. 3 shows an alignment the N-terminal amino acid sequences of mosquito and beetle ADCs described in Examples 5 and 6. N-terminal truncations between the two residues highlighted in black resulted in a truncated ADC having increased activity as compared to its corresponding full-length protein, while N-terminal truncations between the two residues boxed in white resulted in enzymes with low or no detectable ADC activity. The region indicated with a broken line delineates the position where N-terminal truncations may be expected to cease being beneficial for enzymatic activity.

SEQUENCE LISTING

[0034] This application contains a Sequence Listing which has been submitted electronically in computer readable form (ASCII text format) created on Apr. 8, 2024, is named “07 ROBCNZ00700_20240408_sequence_listing.txt”, and is 126,187 bytes in size. The computer readable form is incorporated herein by reference.

TABLE 1

Sequence Listing Description	
SEQ ID NO:	Description
1	<i>Tribolium castaneum</i> (Tc) WT ADC
2	<i>Culex tarsalis</i> (Ct) WT ADC
3	<i>Aethina tumida</i> (At) WT ADC
4	<i>Anopheles arabiensis</i> (Aa) WT ADC
5	<i>Asbolus verrucosus</i> (Av) WT ADC
6	<i>Tenebrio molitor</i> (Tm) WT ADC
7	<i>Cryptotermes secundus</i> (Cs) WT ADC
8	<i>Drosophila melanogaster</i> (Dm) WT ADC
9	<i>Culex quinquefasciatus</i> (Cq) WT ADC
10	<i>Aedes albopictus</i> (Aa2) WT ADC
11	<i>Aedes aegypti</i> (Aa3) WT ADC
12	<i>Anopheles sinensis</i> (As) WT ADC
13	<i>Anopheles albimanus</i> (Aa4) WT ADC

TABLE 1-continued

Sequence Listing Description	
SEQ ID NO:	Description
14	<i>Anopheles darlingi</i> (Ad) WT ADC
15	<i>Anopheles stephensi</i> (As2) WT ADC
16	<i>Tribolium castaneum</i> (Tc) ADC WT cDNA
17	<i>Culex tarsalis</i> (Ct) ADC WT cDNA
18	<i>Aethina tumida</i> (At) ADC WT cDNA
19	<i>Anopheles arabiensis</i> (Aa) ADC WT cDNA
20	<i>Asbolus verrucosus</i> (Av) ADC WT cDNA
21	<i>Tenebrio molitor</i> (Tm) ADC WT cDNA
22	<i>Cryptotermes secundus</i> (Cs) ADC WT cDNA
23	<i>Drosophila melanogaster</i> (Dm) ADC WT cDNA
24	<i>Culex quinquefasciatus</i> (Cq) ADC WT cDNA
25	<i>Aedes albopictus</i> (Aa2) ADC WT cDNA
26	<i>Aedes aegypti</i> (Aa3) ADC WT cDNA
27	<i>Anopheles sinensis</i> (As) ADC WT cDNA
28	Conserved N-term 15-aa mosquito ADC sequence
29	Combined ADC WT sequences sharing greater than 70% identity with N-terminal truncated CtADC

DETAILED DESCRIPTION

[0035] Attempts at industrial-scale biological synthesis of beta-alanine via the enzyme-catalyzed removal of the alpha carboxy group of L-aspartate has been greatly impeded by enzymes having poor activity, expression, and/or stability, thereby making such processes not commercially viable in comparison to chemical synthesis approaches. Improved enzymes catalyzing the conversion of L-aspartate to beta-alanine having increased activity, expression, and/or stability would greatly facilitate the biological synthesis of beta-alanine on a commercial scale. The present description relates to the discovery that certain insect-derived enzymes having aspartate 1-decarboxylase activity are particularly advantageous for beta-alanine production, and further that the performance of such insect-derived enzymes may be greatly improved by truncating portions of their N terminus.

[0036] In a first aspect, described herein are recombinant truncated insect aspartate 1-decarboxylase (ADC) enzymes that are particularly advantageous for beta-alanine production. As used herein, the expression “aspartate 1-decarboxylase” or “ADC” refers to a polypeptide having the ability to catalyze the enzymatic conversion of L-aspartate to beta-alanine and carbon dioxide. In some embodiments, such polypeptides may include those categorized under the enzyme class E.C. 4.1.1.11. In some embodiments, such polypeptides may also include enzymes categorized in other enzyme classes (e.g., enzymes also having activity on substrates other than L-aspartate) and/or polypeptides that may have been annotated (e.g., in public databases) as enzymes other than an ADC (e.g., glutamate decarboxylase, cysteine sulfinic acid decarboxylase). In some embodiments, insect ADCs and truncated variants thereof described herein may include enzymes that have both aspartate 1-decarboxylase activity and cysteine sulfinic acid decarboxylase activity.

[0037] As used herein, the term “truncated” or “truncation” includes not only removal of a segment of a protein starting from a terminal residue (e.g., starting from an N-terminal methionine of a recombinant protein) but may also include deletions of contiguous residues within a terminal region or portion of a protein (e.g., wild-type protein) such that the terminal portion of the truncated protein is shorter than the that of the untruncated protein.

[0038] In some embodiments, the truncated insect ADCs described herein lack a sufficient number of contiguous residues within the amino terminal portion of their corresponding full length wild-type insect ADCs such that the truncated ADC exhibits increased conversion of aspartate to beta-alanine as compared to their parent full length wild-type proteins. In some embodiments, increased conversion of aspartate to beta-alanine may include increased ADC catalytic activity, increased ADC stability, and/or increased expression relative to the corresponding full length wild-type protein.

[0039] In some embodiments, truncated ADCs described herein may be a truncated variant of an organism of the Class Insecta (e.g., a mosquito, fly, beetle, flea, roach, or termite ADC). In particular embodiments, truncated insect ADCs described herein may be a truncated variant of a mosquito, fly, or beetle ADC for which structural relationships are shown in the phylogenetic tree of FIG. 1. In some embodiments, truncated insect ADCs described herein may be a truncated variant of an insect ADC from the genus: *Culex*, *Anopheles*, *Drosophila*, *Aethina*, *Aedes*, *Tribolium*, *Anopheles*, *Tenebrio*, *Asbolus*, or *Cryptotermes*. In some embodiments, truncated insect ADCs described herein may include a truncated variant of an insect ADC from the species: *Culex tarsalis*, *Anopheles arabiensis*, *Drosophila melanogaster*, *Culex quinquefasciatus*, *Aethina tumida*, *Aedes albopictus*, *Aedes aegypti*, *Tribolium castaneum*, *Anopheles sinensis*, *Tenebrio molitor*, *Asbolus verrucosus*, or *Cryptotermes secundus*. In some embodiments, the truncated insect ADCs described herein may be a truncated variant of a wild-type ADC enzyme, the wild-type ADC enzyme being defined by an amino acid sequence at least 90, 91, 92, 93, 94, 95, 96, 97, 98, or 99% identical overall to the amino acid sequence of SEQ ID NO: 29.

[0040] In some embodiments, truncated ADCs described herein may be a truncated variant of a mosquito ADC comprising an amino acid sequence at least 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, or 99% identical overall to any one of SEQ ID NOs: 2, 4, or 9-15. In some embodiments, truncated ADCs described herein may be a truncated variant of a beetle ADC comprising an amino acid sequence at least 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, or 99% identical overall to any one of SEQ ID NOs: 1, 3, or 5-6. In some embodiments, truncated ADCs described herein may be a truncated variant of a fly ADC comprising an amino acid sequence at least 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, or 99% identical overall to SEQ ID NO: 8.

[0041] In some embodiments, truncated ADCs described herein may comprise an amino acid sequence at least 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, or 99% identical overall to an N-terminally truncated fragment of an ADC that exhibits increased activity relative to its untruncated (e.g., full-length) parent enzyme. In some embodiments, truncated ADCs described herein may comprise an amino acid sequence at least 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, or 99% identical overall to: (a) position 72 to 561 of the amino acid sequence of CtADC set forth in SEQ ID NO: 2; (b) position 79 to 568 of the amino acid sequence of AaADC set forth in SEQ ID NO: 4; (c) position 56 to 544 of the amino acid sequence of AtADC set forth in SEQ ID NO: 3; (d) position 52 to 540 of the amino acid sequence of TcADC set forth in

SEQ ID NO: 1; (e) position 71 to 560 of the amino acid sequence of Aa2ADC set forth in SEQ ID NO: 10; (f) position 71 to 562 of the amino acid sequence of Aa3ADC set forth in SEQ ID NO: 11; (g) position 74 to 563 of the amino acid sequence of CqADC set forth in SEQ ID NO: 9; (h) position 72 to 561 of the amino acid sequence of Aa4ADC set forth in SEQ ID NO: 13; (i) position 74 to 624 of the amino acid sequence of AdADC set forth in SEQ ID NO: 14; (j) position 83 to 572 of the amino acid sequence of AsADC set forth in SEQ ID NO: 12; (k) position 72 to 561 of the amino acid sequence of As2ADC set forth in SEQ ID NO: 15; (l) position 53 to 541 of the amino acid sequence of TmADC set forth in SEQ ID NO: 6; (m) position 57 to 572 of the amino acid sequence of AvADC set forth in SEQ ID NO: 5; or (n) position 1 to 491 the amino acid sequence set forth in SEQ ID NO: 29. These segments correspond to fragments of wild-type full length insect ADCs that either are shown to exhibit increased performance in beta-alanine production or may be expected to do so based on sequence conservation and multiple sequence alignments such as in Examples 5-7 and in FIGS. 2 and 3.

[0042] In some embodiments, truncated ADCs described herein may lack at least X contiguous residues of the amino terminus of the corresponding full length wild-type insect ADC, wherein X is any integer between 5 and 50. In some embodiments, truncated ADCs described herein may lack at least 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, or 70 contiguous residues of the amino terminus of the corresponding full length wild-type insect ADC, depending on the length of the amino terminus of the corresponding full length wild-type insect ADC.

[0043] In some embodiments, truncated ADCs described herein may be truncated at a position immediately C-terminal (downstream) of a residue corresponding to position n of a full-length wild-type insect ADC, wherein n is any integer between 2 and Y, wherein Y is the most C-terminal residue position within the full-length wild-type insect ADC at which truncation can occur with the truncated ADC exhibiting increased conversion of aspartate to beta-alanine as compared to the full length wild-type ADC. As used herein in the context of amino acid residue numbering, the expression "corresponding to position" considers that amino acid residue numbering differs amongst different proteins (e.g., different insect ADCs) but that a person of skill in the art would be able to determine corresponding residue positions in two proteins sharing a degree of amino acid sequence identity by performing sequence alignments between the two proteins, optionally including additional orthologs to identify conserved residues using widely available softwares (e.g., Clustal Omega) as demonstrated herein.

[0044] In some embodiments, truncated ADCs described herein may be truncated at a position C-terminal (downstream) of a residue corresponding to any one of: (a) position 2, 3, 4, 5, 6, 7, 8, 9, 10, or 11 to 71 of the amino acid sequence of CtADC set forth in SEQ ID NO: 2; (b) position 2, 3, 4, 5, 6, 7, 8, 9, 10, or 11 to 78 of the amino acid sequence of AaADC set forth in SEQ ID NO: 4; (c) position 2, 3, 4, 5, 6, 7, 8, 9, 10, or 11 to 55 of the amino acid sequence of AtADC set forth in SEQ ID NO: 3; (d) position 2, 3, 4, 5, 6, 7, 8, 9, 10, or 11 to 51 of the amino acid sequence of TcADC set forth in SEQ ID NO: 1; (e) position 2, 3, 4, 5,

6, 7, 8, 9 10, or 11 to 70 of the amino acid sequence of Aa2ADC set forth in SEQ ID NO: 10; (f) position 2, 3, 4, 5, 6, 7, 8, 9 10, or 11 to 70 of the amino acid sequence of Aa3ADC set forth in SEQ ID NO: 11; (g) position 2, 3, 4, 5, 6, 7, 8, 9 10, or 11 to 73 of the amino acid sequence of CqADC set forth in SEQ ID NO: 9; (h) position 2, 3, 4, 5, 6, 7, 8, 9 10, or 11 to 73 of the amino acid sequence of Aa4ADC set forth in SEQ ID NO: 13; (i) position 2, 3, 4, 5, 6, 7, 8, 9 10, or 11 to 73 of the amino acid sequence of AdADC set forth in SEQ ID NO: 14; (j) position 2, 3, 4, 5, 6, 7, 8, 9 10, or 11 to 82 of the amino acid sequence of AsADC set forth in SEQ ID NO: 12; (k) position 2, 3, 4, 5, 6, 7, 8, 9 10, or 11 to 78 of the amino acid sequence of As2ADC set forth in SEQ ID NO: 15; (l) position 2, 3, 4, 5, 6, 7, 8, 9 10, or 11 to 52 of the amino acid sequence of TmADC set forth in SEQ ID NO: 6; (m) position 2, 3, 4, 5, 6, 7, 8, 9 10, or 11 to 56 of the amino acid sequence of AvADC set forth in SEQ ID NO: 5; or (n) position 1 to 491 of the amino acid consensus sequence of SEQ ID NO: 29. The upper limit of each of the amino acid sequences mentioned above refers to the residue positions within the full length wild-type insect ADCs shown in Table 7 and FIG. 3 and corresponding to K71 of CtADC. N-terminal truncations at least up to K71 of CtADC resulted in a truncated ADC (CtADC72-561) exhibiting increased conversion of aspartate to beta-alanine as compared to the full length wild-type ADC.

[0045] In some embodiments, truncated ADCs described herein may be truncated at a position N-terminal (upstream) of a residue corresponding to any one of: (a) position 72 to 80 of the amino acid sequence of CtADC set forth in SEQ ID NO: 2; (b) position 79 to 87 of the amino acid sequence of AaADC set forth in SEQ ID NO: 4; (c) position 56 to 64 of the amino acid sequence of AtADC set forth in SEQ ID NO: 3; (d) position 52 to 60 of the amino acid sequence of TcADC set forth in SEQ ID NO: 1; (e) position 71 to 79 of the amino acid sequence of Aa2ADC set forth in SEQ ID NO: 10; (f) position 71 to 79 of the amino acid sequence of Aa3ADC set forth in SEQ ID NO: 11; (g) position 74 to 82 of the amino acid sequence of CqADC set forth in SEQ ID NO: 9; (h) position 72 to 82 of the amino acid sequence of Aa4ADC set forth in SEQ ID NO: 13; (i) position 74 to 82 of the amino acid sequence of AdADC set forth in SEQ ID NO: 14; (j) position 83 to 91 of the amino acid sequence of AsADC set forth in SEQ ID NO: 12; (k) position 72 to 87 of the amino acid sequence of As2ADC set forth in SEQ ID NO: 15; (l) position 53 to 61 of the amino acid sequence of TmADC set forth in SEQ ID NO: 6; (m) position 57 to 65 of the amino acid sequence of AvADC set forth in SEQ ID NO: 5; or (n) position 1 to 491 the amino acid sequence set forth in SEQ ID NO: 29. These residue positions correspond to those delineated in Table 7 and by a broken line in FIG. 3.

[0046] In some embodiments, truncated ADCs described herein may be truncated at a position N-terminal (upstream) of a residue corresponding to any one of: (a) position 75 of the amino acid sequence of CtADC set forth in SEQ ID NO: 2; (b) position 82 of the amino acid sequence of AaADC set forth in SEQ ID NO: 4; (c) position 59 of the amino acid sequence of AtADC set forth in SEQ ID NO: 3; (d) position 55 of the amino acid sequence of TcADC set forth in SEQ ID NO: 1; (e) position 74 of the amino acid sequence of Aa2ADC set forth in SEQ ID NO: 10; (f) position 74 of the amino acid sequence of Aa3ADC set forth in SEQ ID NO:

11; (g) position 77 of the amino acid sequence of CqADC set forth in SEQ ID NO: 9; (h) position 77 of the amino acid sequence of Aa4ADC set forth in SEQ ID NO: 13; (i) position 77 of the amino acid sequence of AdADC set forth in SEQ ID NO: 14; (j) position 86 of the amino acid sequence of AsADC set forth in SEQ ID NO: 12; (k) position 82 of the amino acid sequence of As2ADC set forth in SEQ ID NO: 15; (l) position 56 of the amino acid sequence of TmADC set forth in SEQ ID NO: 6; (m) position 60 of the amino acid sequence of AvADC set forth in SEQ ID NO: 5; or (n) position 1 to 491 the amino acid sequence set forth in SEQ ID NO: 29. These residue positions correspond to S75 in CtADC that occurs within a tripeptide sequence “SLP” being conserved across all the insect sequences aligned in FIG. 3.

[0047] In a further aspect, described herein are recombinant proteins having aspartate 1-decarboxylase activity, the recombinant proteins comprising an amino acid sequence at least 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, or 99% identical overall to: (a) position 72 to 561 of the amino acid sequence of CtADC set forth in SEQ ID NO: 2; (b) position 79 to 568 of the amino acid sequence of AaADC set forth in SEQ ID NO: 4; (c) position 56 to 544 of the amino acid sequence of AtADC set forth in SEQ ID NO: 3; (d) position 52 to 540 of the amino acid sequence of TcADC set forth in SEQ ID NO: 1; (e) position 71 to 560 of the amino acid sequence of Aa2ADC set forth in SEQ ID NO: 10; (f) position 71 to 562 of the amino acid sequence of Aa3ADC set forth in SEQ ID NO: 11; (g) position 74 to 563 of the amino acid sequence of CqADC set forth in SEQ ID NO: 9; (h) position 72 to 561 of the amino acid sequence of Aa4ADC set forth in SEQ ID NO: 13; (i) position 74 to 624 of the amino acid sequence of AdADC set forth in SEQ ID NO: 14; (j) position 83 to 572 of the amino acid sequence of AsADC set forth in SEQ ID NO: 12; (k) position 72 to 561 of the amino acid sequence of As2ADC set forth in SEQ ID NO: 15; (l) position 53 to 541 of the amino acid sequence of TmADC set forth in SEQ ID NO: 6; (m) position 57 to 572 of the amino acid sequence of AvADC set forth in SEQ ID NO: 5; or (n) position 1 to 491 the amino acid sequence set forth in SEQ ID NO: 29. These regions were not only found herein to be highly conserved amongst at least mosquito and beetle ADCs, but they also correspond to a truncated variant of CtADC72-561 which was found to exhibit increased conversion of aspartate to beta-alanine as compared to the full length wild-type CtADC.

[0048] In some embodiments, truncated ADCs and/or recombinant proteins described herein may comprise a glycine residue at a position corresponding to position 96 of the amino acid sequence of CtADC set forth in SEQ ID NO: 2. CtADC outperformed other insect-derived ADCs by a significant margin from the enzyme activity testing performed in Example 2, including a 39% increase in activity over counterpart mosquito enzyme CqADC with which it shares about 97% amino acid sequence identity. A comparison of amino acid differences between CtADC and CqADC in the catalytic portion of the enzymes performed in Example 8 revealed a single glycine residue at position 96 of CtADC that was unique amongst all other insect sequences analyzed (see FIG. 3), suggesting that this residue may play a role in the heightened beta-alanine production associated with CtADC.

[0049] In a further aspect, described herein is a polynucleotide comprising a nucleic acid sequence encoding a recom-

binant truncated insect ADC or a recombinant protein as described herein. In some embodiments, the polynucleotide is DNA. In some embodiments, the polynucleotide is RNA.

[0050] In a further aspect, described herein is a polynucleotide molecule that hybridizes under stringent conditions to the full complement of the nucleic acid sequence of any one of SEQ ID NOs: 16 to 27; a nucleic acid sequence encoding the recombinant truncated insect ADC as described herein; a nucleic acid sequence encoding the truncated recombinant protein as described herein; or any combination thereof.

[0051] In a further aspect, described herein is an expression cassette comprising the isolated or recombinant polynucleotide described herein operably linked to a promoter (e.g., that is heterologous with respect to the insect ADC).

[0052] In a further aspect, described herein is a host cell that expresses a recombinant truncated insect ADC or a recombinant protein as described herein, and/or is transformed with or engineered to comprise a polynucleotide or expression cassette described herein. In some embodiments, the host cell may be a microbial cell. In some embodiments, the host cell may be a bacterial, insect, mammalian, yeast or fungal cell.

[0053] In a further aspect, a recombinant truncated insect ADC, a recombinant protein, or a host cell described herein, may be for use in the industrial production of beta-alanine from aspartate. In a further aspect, described herein is a process for the production of beta-alanine, the process comprising: (a) providing an ADC enzyme source which is a truncated insect ADC as described herein, a recombinant protein as described herein, and/or a host cell as described herein; (b) contacting the ADC enzyme source with a source of aspartate under conditions enabling the enzyme source to catalyze the conversion of the aspartate to beta-alanine; and (c) isolating and/or concentrating the beta-alanine produced. In some embodiments, the host cells expressing the recombinant truncated insect ADC or the recombinant protein described herein may be utilized as intact cells, which advantageously may prevent cellular debris from lysed cells from contaminating the beta-alanine produced.

[0054] In a further aspect, described herein is a composition comprising beta-alanine produced by a process described herein.

EXAMPLES

Example 1: General Materials and Methods

Cloning and Expression of L-Aspartate- α -Decarboxylase (ADC) Enzymes

[0055] Codon optimized cDNA sequences of the ADCs that were cloned and expressed in bacteria are shown in SEQ ID NOs: 16-27. The cDNA sequences of ADCs were cloned into separate expression vectors and transformed into *Escherichia coli* to enhance the expression of ADCs upon addition of an inducer. For N-terminal truncations, the desired number of amino acids downstream of the initiator methionine were deleted.

ADC Activity Measurements

[0056] ADC activity was measured by first, growing BL21 (DE3) *E. coli* cells expressing the ADC of interest in 500 μ L LB broth containing kanamycin and 0.2% isopropyl β -D-thiogalactopyranoside (IPTG) for 24 hours at 30° C. Cells were then pelleted to remove the supernatant, resuspended

and ultrasonic-crushed. The plate was then centrifuged to remove any debris, and the supernatant containing the cell lysates was collected. The cell lysates containing the ADCs were then tested for activity by incubating 50 μ L of the supernatant in a 50 mL solution containing L-aspartic acid with a final concentration of 60 g/L and pyridoxal phosphate (PLP) with a final concentration of 0.2 g/L at a pH of 6.5, a temperature of 37° C., and stirred at 200 rpm. 1M of sulfuric acid was then titrated into the reaction solution to maintain the pH. After one hour, the amount of sulfuric acid used by the reaction was determined to directly measure ADC activity. Experiments were conducted at least 3 times, and the average activity value was calculated.

Example 2: Activity of Insect-Derived ADCs

[0057] A large-scale screening was performed to compare the expression and activity of ADC enzymes from a plurality of different prokaryotic and eukaryotic organisms when recombinantly expressed in bacterial host cells. The screening revealed that lysates of bacterial cells transformed with ADCs from insect species exhibited consistently higher beta-alanine production than ADCs from other organisms. Table 2 shows relative ADC activities of lysates from bacteria transformed with codon-optimized cDNAs of ADCs from mosquito, fly and beetle species, measured as described in Example 1. Interestingly, lysate from bacteria transformed with an ADC from the mosquito species *Culex tarsalis* (CtADC; SEQ ID NO: 2) significantly outperformed all other enzymes tested.

TABLE 2

Activity of ADCs				
Enzyme	SEQ ID NO:	Activity	Insect	Species
CtADC	2	2.5	mosquito	<i>Culex tarsalis</i>
AaADC	4	2.0	mosquito	<i>Anopheles arabiensis</i>
DmADC	8	2.0	fly	<i>Drosophila melanogaster</i>
CqADC	9	1.8	mosquito	<i>Culex quinquefasciatus</i>
AtADC	3	1.6	beetle	<i>Aethina tumida</i>
Aa2ADC	10	1.5	mosquito	<i>Aedes albopictus</i>
Aa3ADC	11	1.25	mosquito	<i>Aedes aegypti</i>
TcADC	1	0.7	beetle	<i>Tribolium castaneum</i>
AsADC	12	1.0	mosquito	<i>Anopheles sinensis</i>
TmADC	6	0.6	beetle	<i>Tenebrio molitor</i>
AvADC	5	0.1	beetle	<i>Asbolus verrucosus</i>

Example 3: Insect-Derived ADC Sequence Analysis

[0058] The amino acid sequence of CtADC was used as the basis of a Protein BLAST™ to identify other ADCs from various species. Over 5000 hits sequences were retrieved, of which 188 with the highest BLAST score sequences were then selected, combined with the sequences of the insect-derived ADCs of Table 2, grouped by 85% sequence identity, and finally incorporated in a broad insect phylogenetic tree (FIG. 1). The phylogenetic tree shown in FIG. 1 shows that mosquito and fly ADCs are structurally related, and that the beetle, flea, roach and termite ADCs are structurally related.

Example 4: Mosquito-Derived ADC Sequence Analysis

[0059] An alignment of amino acid sequences of ADCs identified from nine different mosquito species was performed with Clustal Omega (1.2.4) and is shown in FIG. 2. The alignment revealed relatively high sequence conservation throughout different mosquito species, as shown below in the percent identity matrix in Table 3.

TABLE 3

Percent identity matrix of mosquito-derived ADCs									
	Aa2ADC	Aa3ADC	CtADC	CqADC	Aa4ADC	AdADC	AaADC	AsADC	As2ADC
Aa2ADC	100.00	97.86	89.11	88.75	82.52	82.52	84.74	86.58	85.13
Aa3ADC	97.86	100.00	88.93	88.57	81.98	82.16	84.92	86.23	84.41
CtADC	89.11	88.93	100.00	97.33	81.12	81.12	82.59	84.11	83.36
CqADC	88.75	88.57	97.33	100.00	80.94	80.94	82.29	83.99	83.54
Aa4ADC	82.52	81.98	81.12	80.94	100.00	98.05	89.07	89.17	89.70
AdADC	82.52	82.16	81.12	80.94	98.05	100.00	89.43	88.63	89.88
AaADC	84.74	84.92	82.59	82.29	89.07	89.43	100.00	92.06	92.72
AsADC	86.58	86.23	84.11	83.99	89.17	88.63	92.06	100.00	92.59
As2ADC	85.13	84.41	83.36	83.54	89.70	89.88	92.72	92.59	100.00

Example 5: N-Terminal Truncations of Mosquito ADCs Resulted in Higher Beta-Alanine Production

[0060] Interestingly, the alignment in FIG. 2 revealed a region of low sequence conservation towards the amino terminus of the mosquito ADCs indicated in a broken line in FIG. 2, which immediately followed a 15-amino acid segment (SGSDSAGVSEDEDVQ; SEQ ID NO: 28) that was 100% conserved across all mosquito ADCs analyzed. To study the role of the N-terminus of CtADC in its activity, progressive N-terminal truncations were generated and expressed in bacteria, and their ADC activities were characterized as described in Example 1. Strikingly, N-terminal truncations ranging from 11 to 71 amino acids increased beta-alanine production by 24% to 100% as shown in Table 4. However, no ADC enzymatic activity was detected by truncating 81 amino acids or more from the N terminus of CtADC.

TABLE 4

Activities of N-terminal truncations of CtADC		
CtADC Truncation	AA sequence relative to SEQ ID NO: 2	Activity
CtADC (full length)	1-561	2.5
CtADCN1	12-561	3.55
CtADCN2	22-561	4.2
CtADCN3	32-561	4.95
CtADCN4	42-561	4.8
CtADCN5	52-561	3.4
CtADCN6	62-561	4.8
CtADCN7	72-561	3.1
CtADCN8	82-561	/
CtADCN9	92-561	/
CtADCN10	102-561	/
CtADCN11	131-561	/
CtADCN12	171-561	/
CtADCN13	201-561	/

“/”: Activity was too low to detect.

[0061] N-terminal truncations were also generated and characterized for another mosquito enzyme, AaADC as shown in Table 5. A 70% increase in beta-alanine production was observed by truncating the N-terminal 63 amino acids of AaADC. However, no ADC enzymatic activity was detected by truncating 137 amino acids or more of AaADC.

TABLE 5

Activities of N-terminal truncations of AaADCs		
AaADC Truncation	AA sequence relative to SEQ ID NO: 4	Activity
AaADC (full-length)	1-561	2.0
AaADCN1	64-568	3.4
AaADCN2	138-568	/
AaADCN3	208-568	/

“/”: Activity was too low to detect.

Example 6: N-Terminal Truncations of Beetle ADCs Resulted in Higher Beta-Alanine Production

[0062] Progressive N-terminal truncations were generated and expressed in bacteria for two beetle ADCs and their ADC activities were characterized as described in Example 1. Results are shown in Tables 5 and 6 for AtADC and TcADC. For AtADC, a striking 256% increase in beta-alanine production was observed by truncating the N-terminal 45 amino acids. However, no ADC enzymatic activity was detected by truncating 114 amino acids or more of AtADC (Table 5). For TcADC, N-terminal truncations ranging from 10 to 50 amino acids increased beta-alanine production by 10% to 330%. However, no ADC enzymatic activity was detected by truncating 60 amino acids from the N terminus of TcADC (TcADCN6, Table 6).

TABLE 5

Activities of truncated AtADCs		
AtADC Truncation	AA sequence relative to SEQ ID NO: 3	Activity
AtADC (full length)	1-544	1.6
AtADCN1	46-544	4.1
AtADCN2	115-544	/
AtADCN3	185-544	/

“/”: Activity was too low to detect.

TABLE 6

Activities of truncated TcADCs		
TcADC Truncation	AA sequence relative to SEQ ID NO: 1	Activity
TcADC (full-length)	1-540	1.0
TcADCN1	11-540	2.8

TABLE 6-continued

Activities of truncated TcADCs		
TcADC Truncation	AA sequence relative to SEQ ID NO: 1	Activity
TcADCN2	21-540	1.1
TcADCN3	31-540	1.25
TcADCN4	41-540	3.3
TcADCN5	51-540	2.3
TcADCN6	61-540	/

"/": Activity was too low to detect.

Example 7: Analysis of Positions of N-Terminal Truncations Resulting in Higher Beta-Alanine Production

[0063] An alignment of the N-terminal sequences of the mosquito and beetle ADCs described in Examples 5 and 6 is shown in FIG. 3. The alignment in FIG. 3 helps visualize and understand the N-terminal truncation results in Examples 5 and 6, in which an N-terminal truncation between the two residues highlighted in black resulted in a truncated ADC having increased activity as compared to its corresponding full-length protein. Conversely, an N-terminal truncation between the two residues boxed in white resulted in a truncated ADC having no detectable ADC activity. Thus, with the truncation experiments for CtADC and TcADC providing the highest resolution, the area indicated with a broken line delineates the position where an N-terminal truncation is expected to cease being beneficial for beta-alanine production. The respective positions of the residues in the mosquito and beetle ADCs are shown in Table 7. The area marked with a broken line in FIG. 3 also coincides with the start of greater sequence conservation across the mosquito and beetle ADCs, with a tripeptide sequence “SLP” being 100% conserved across all the sequences aligned. Without wishing to be bound by theory, truncations N-terminal to (or upstream of) the serine within the conserved “SLP” tripeptide may be beneficial for increased beta-alanine production, while truncations downstream of the conserved serine may be detrimental (Table 7).

TABLE 7

Residue positions indicated in FIG. 3			
ADC	SEQ ID NO:	Position of area marked in a broken line in FIG. 3	Position of start of conserved SLP tripeptide
CtADC	2	R72-E80	S75
AaADC	4	R79-E87	S82

TABLE 7-continued

Residue positions indicated in FIG. 3			
ADC	SEQ ID NO:	Position of area marked in a broken line in FIG. 3	Position of start of conserved SLP tripeptide
AtADC	3	T56-A64	S59
TcADC	1	S52-R60	S55
Aa2ADC	10	R71-E79	S74
Aa3ADC	11	R71-E79	S74
CqADC	9	R74-E82	S77
Aa4ADC	13	R74-E82	S77
AdADC	14	R74-E82	S77
AsADC	12	R83-E91	S86
As2ADC	15	R79-E87	S82
TmADC	6	S53-E61	S56
AvADC	5	S57-E65	S60

[0064] The N-terminal truncated CtADC sequence beginning at S75 was used as the basis for a further Protein BLAST™ search to identify other ADCs from various species. The ADC ortholog sequences sharing at least 70% amino acid identity with the truncated CtADC sequence were subjected to multiple sequence alignment analyses and a consensus ADC sequence is shown in SEQ ID NO: 29.

Example 8: Comparison of CtADC with ADCs from Other Mosquito Species

[0065] CtADC outperformed other insect-derived ADCs by a significant margin from the enzyme activity testing performed in Example 2. Based on the activities shown in Table 2, CtADC exhibited a 25% increase in beta-alanine production over the next best insect-derived ADCs from mosquito (AaADCs) and fly (DmADC). Interestingly, CtADC shares about 97% overall amino acid sequence identity with CqADC (which is also derived from mosquito) yet the results in Table 2 reveal that CtADC exhibited 39% higher beta-alanine production than CqADC. The results shown in Table 4 reveal that at least the N-terminal 71 residues of CtADC may be truncated without abrogating ADC activity (CtADCN7). Thus, looking at amino acid differences between CtADC and CqADC within residues 72-561 of CtADC revealed only seven amino acid substitutions. Six of the seven amino acid substitutions correspond to residues that are found in different mosquito ADC orthologs. Interestingly, the only residue that was unique to CtADC was a glycine at position 96 (see residue highlighted in black in FIG. 2). In fact, a glycine corresponding to position 96 of the full-length CtADC (SEQ ID NO: 2) was not found in any other mosquito or beetle sequence analyzed (see FIG. 3), suggesting that this residue may play a role in the heightened beta-alanine production associated with CtADC.

SEQUENCE LISTING

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<210> SEQ ID NO 1
<211> LENGTH: 540
<212> TYPE: PRT
<213> ORGANISM: Tribolium castaneum
<400> SEQUENCE: 1
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Met	Pro	Ala	Thr	Gly	Glu	Asp	Gln	Asp	Leu	Val	Gln	Asp	Leu	Ile	Glu	1	5	10	15
Glu	Pro	Ala	Thr	Phe	Ser	Asp	Ala	Val	Leu	Ser	Ser	Asp	Glu	Glu	Leu	20	25	30	
Phe	His	Gln	Lys	Cys	Pro	Lys	Pro	Ala	Pro	Ile	Tyr	Ser	Pro	Val	Ser	35	40	45	
Lys	Pro	Val	Ser	Phe	Glu	Ser	Leu	Pro	Asn	Arg	Arg	Leu	His	Glu	Glu	50	55	60	
Phe	Leu	Arg	Ser	Ser	Val	Asp	Val	Leu	Leu	Gln	Glu	Ala	Val	Phe	Glu	65	70	75	80
Gly	Thr	Asn	Arg	Lys	Asn	Arg	Val	Leu	Gln	Trp	Arg	Glu	Pro	Glu	Glu	85	90	95	
Leu	Arg	Arg	Leu	Met	Asp	Phe	Gly	Val	Arg	Ser	Ala	Pro	Ser	Thr	His	100	105	110	
Glu	Glu	Leu	Leu	Glu	Val	Leu	Lys	Lys	Val	Val	Thr	Tyr	Ser	Val	Lys	115	120	125	
Thr	Gly	His	Pro	Tyr	Phe	Val	Asn	Gln	Leu	Phe	Ser	Ala	Val	Asp	Pro	130	135	140	
Tyr	Gly	Leu	Val	Ala	Gln	Trp	Ala	Thr	Asp	Ala	Leu	Asn	Pro	Ser	Val	145	150	155	160
Tyr	Thr	Tyr	Glu	Val	Ser	Pro	Val	Phe	Val	Leu	Met	Glu	Glu	Val	Val	165	170	175	
Leu	Arg	Glu	Met	Arg	Ala	Ile	Val	Gly	Phe	Glu	Gly	Gly	Lys	Gly	Asp	180	185	190	
Gly	Ile	Phe	Cys	Pro	Gly	Gly	Ser	Ile	Ala	Asn	Gly	Tyr	Ala	Ile	Ser	195	200	205	
Cys	Ala	Arg	Tyr	Arg	Phe	Met	Pro	Asp	Ile	Lys	Lys	Lys	Gly	Leu	His	210	215	220	
Ser	Leu	Pro	Arg	Leu	Val	Leu	Phe	Thr	Ser	Glu	Asp	Ala	His	Tyr	Ser	225	230	235	240
Ile	Lys	Lys	Leu	Ala	Ser	Phe	Gln	Gly	Ile	Gly	Thr	Asp	Asn	Val	Tyr	245	250	255	
Leu	Ile	Arg	Thr	Asp	Ala	Arg	Gly	Arg	Met	Asp	Val	Ser	His	Leu	Val	260	265	270	
Glu	Glu	Ile	Glu	Arg	Ser	Leu	Arg	Glu	Gly	Ala	Ala	Pro	Phe	Met	Val	275	280	285	
Ser	Ala	Thr	Ala	Gly	Thr	Thr	Val	Ile	Gly	Ala	Phe	Asp	Pro	Ile	Glu	290	295	300	
Lys	Ile	Ala	Asp	Val	Cys	Gln	Lys	Tyr	Lys	Leu	Trp	Leu	His	Val	Asp	305	310	315	320
Ala	Ala	Trp	Gly	Gly	Gly	Ala	Leu	Val	Ser	Ala	Lys	His	Arg	His	Leu	325	330	335	
Leu	Lys	Gly	Ile	Glu	Arg	Ala	Asp	Ser	Val	Thr	Trp	Asn	Pro	His	Lys	340	345	350	
Leu	Leu	Thr	Ala	Pro	Gln	Gln	Cys	Ser	Thr	Leu	Leu	Leu	Arg	His	Glu	355	360	365	
Gly	Val	Leu	Ala	Glu	Ala	His	Ser	Thr	Asn	Ala	Ala	Tyr	Leu	Phe	Gln	370	375	380	
Lys	Asp	Lys	Phe	Tyr	Asp	Thr	Lys	Tyr	Asp	Thr	Gly	Asp	Lys	His	Ile	385	390	395	400
Gln	Cys	Gly	Arg	Arg	Ala	Asp	Val	Leu	Lys	Phe	Trp	Phe	Met	Trp	Lys				

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Ala Lys Gly Thr Ser Gly Leu Glu Lys His Val Asp Lys Val Phe Glu	405	410	415
	420	425	430
Asn Ala Arg Phe Phe Thr Asp Cys Ile Lys Asn Arg Glu Gly Phe Glu	435	440	445
Met Val Ile Ala Glu Pro Glu Tyr Thr Asn Ile Cys Phe Trp Tyr Val	450	455	460
Pro Lys Ser Leu Arg Gly Arg Lys Asp Glu Ala Asp Tyr Lys Asp Lys	465	470	475
			480
Leu His Lys Val Ala Pro Arg Ile Lys Glu Arg Met Met Lys Glu Gly	485	490	495
Ser Met Met Val Thr Tyr Gln Ala Gln Lys Gly His Pro Asn Phe Phe	500	505	510
Arg Ile Val Phe Gln Asn Ser Gly Leu Asp Lys Ala Asp Met Val His	515	520	525
Leu Val Glu Glu Ile Glu Arg Leu Gly Ser Asp Leu	530	535	540

<210> SEQ ID NO 2

<211> LENGTH: 561

<212> TYPE: PRT

<213> ORGANISM: Culex tarsalis

<400> SEQUENCE: 2

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Ala Asn Leu Ser Ser Gly Ser Asp Ser Ala Gly Val Ser Glu Asp Glu	20	25	30	
Asp Val Gln Leu Phe Ser Thr Thr Gly Asn Ile Val Ser Ser Lys Pro	35	40	45	
Leu Lys Lys Pro Ala Leu Lys Pro Ala Thr Lys Asp Glu Asp Gln Asn	50	55	60	
Lys Thr Lys Ala Asn Ala Lys Arg Tyr Ala Ser Leu Pro Asn Arg Glu	65	70	75	80
Gln His Gln Arg Phe Leu Thr Asp Phe Leu Ser Glu Val Leu Asn Gly	85	90	95	
Ala Ile Phe Asn Ala Thr Asp Arg Ser Asn Lys Val Leu Asn Trp Val	100	105	110	
Asp Pro Glu Glu Leu Lys Arg Ser Ile Asp Leu Ser Leu Lys Asp Glu	115	120	125	
Pro Asp Ser Asp Glu Lys Leu Leu Glu Leu Ala Arg Ala Thr Ile Asp	130	135	140	
His Ser Val Lys Thr Gly His Pro Tyr Phe Met Asn Gln Leu Phe Ser	145	150	155	160
Ser Val Asp Pro Tyr Gly Phe Ala Gly Gln Val Leu Thr Asp Ala Leu	165	170	175	
Asn Pro Ser Val Tyr Thr Phe Glu Val Ser Pro Val Phe Val Leu Met	180	185	190	
Glu Glu Val Val Leu Lys Glu Met Arg Thr Ile Val Gly Phe Pro Gly	195	200	205	
Gly Val Gly Asp Gly Ile Phe Cys Pro Gly Gly Ser Met Ala Asn Gly	210	215	220	

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Tyr Ala Ile Ser Cys Ala Arg Phe Lys His Met Pro Asp Val Lys Thr
225                230                235                240

Lys Gly Leu His Ser Leu Pro Arg Leu Val Ile Phe Thr Ser Glu Asp
                245                250                255

Ala His Tyr Ser Ile Lys Lys Leu Ala Ser Phe Met Gly Ile Gly Ser
                260                265                270

Asp Asn Val Tyr Pro Ile Arg Thr Asp Ala Val Gly Lys Ile Gln Pro
                275                280                285

Asp His Leu Glu Ala Glu Ile Leu Arg Ala Lys Ser Glu Gly Ala Val
                290                295                300

Pro Phe Met Val Ser Ala Thr Ala Gly Thr Thr Val Ile Gly Ala Phe
305                310                315                320

Asp Pro Leu Glu Gln Ile Ala Asp Leu Cys Gln Lys Tyr Asn Leu Trp
                325                330                335

Met His Val Asp Ala Ala Trp Gly Gly Gly Ala Leu Met Ser Lys Lys
                340                345                350

Tyr Arg Thr Leu Leu Lys Gly Val Glu Arg Ala Asp Ser Val Thr Trp
                355                360                365

Asn Pro His Lys Leu Leu Ala Ala Pro Gln Gln Cys Ser Thr Phe Leu
                370                375                380

Thr Arg His Glu Gly Ile Leu Ser Gly Cys His Ser Thr Asn Ala Thr
385                390                395                400

Tyr Leu Phe Gln Lys Asp Lys Phe Tyr Asp Thr Gln Tyr Asp Thr Gly
                405                410                415

Asp Lys His Ile Gln Cys Gly Arg Arg Ala Asp Val Leu Lys Phe Trp
                420                425                430

Phe Met Trp Arg Ala Lys Gly Thr Ser Gly Leu Glu Gln His Ile Asp
                435                440                445

Lys Val Phe Glu Thr Ala Glu Tyr Phe Thr Asn Ser Ile Lys Ala Arg
                450                455                460

Pro Gly Phe Glu Met Val Ile Glu Asn Pro Glu Cys Thr Asn Val Cys
465                470                475                480

Phe Trp Tyr Val Pro Pro Gly Leu Arg Gln Val Pro Arg Asp Ser Ala
                485                490                495

Glu Phe Gly Glu Arg Leu His Lys Val Ala Pro Lys Val Lys Glu Arg
                500                505                510

Met Met Arg Glu Gly Ser Met Met Ile Thr Tyr Gln Pro Ile His Asp
                515                520                525

Lys Pro Asn Phe Phe Arg Leu Val Leu Gln Asn Ser Ala Leu Asp Lys
                530                535                540

Ser Asp Met Asn Tyr Ile Ile Asp Glu Ile Glu Arg Leu Ala Ser Asp
545                550                555                560

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Leu

<210> SEQ ID NO 3

<211> LENGTH: 544

<212> TYPE: PRT

<213> ORGANISM: Aethina tumida

<400> SEQUENCE: 3

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1              5              10              15

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			20				25						30			
Val	Gln	Tyr	Ser	Asn	Asp	Glu	Arg	Ser	Ile	Arg	Asp	Met	Lys	Ala	Thr	
			35				40				45					
Ile	Ala	Thr	Gly	Lys	Leu	Ala	Thr	Phe	Glu	Ser	Leu	Pro	Ser	Arg	Ala	
			50	55						60						
His	His	Glu	Glu	Phe	Ile	Arg	Ser	Cys	Met	Asp	Val	Ile	Leu	Lys	Glu	
65				70						75				80		
Ala	Val	Phe	Asp	Gly	Thr	Asn	Arg	Asn	Asn	Pro	Val	Leu	Asn	Phe	Val	
			85				90						95			
Asn	Pro	Glu	Glu	Leu	Gln	Ser	Lys	Val	Asn	Phe	Lys	Leu	Lys	Thr	Ala	
			100				105				110					
Pro	Ser	Thr	His	Glu	Asp	Leu	Leu	Lys	Thr	Leu	Lys	Asp	Thr	Ile	Arg	
			115	120						125						
Tyr	Ser	Val	Lys	Thr	Gly	His	Pro	Tyr	Phe	Val	Asn	Gln	Leu	Phe	Ser	
			130	135						140						
Ser	Leu	Asp	Pro	Tyr	Gly	Leu	Val	Gly	Gln	Trp	Leu	Thr	Asp	Ala	Leu	
145				150						155				160		
Asn	Pro	Thr	Val	Tyr	Thr	Tyr	Glu	Val	Ser	Pro	Val	Phe	Thr	Leu	Met	
			165				170						175			
Glu	Glu	Glu	Val	Leu	Arg	Glu	Met	Arg	Thr	Ile	Val	Gly	Phe	Lys	Asn	
			180				185				190					
Gly	Glu	Gly	Asp	Gly	Ile	Phe	Cys	Pro	Gly	Gly	Ser	Met	Ala	Asn	Gly	
			195	200						205						
Tyr	Ala	Ile	Ser	Cys	Ala	Arg	His	Lys	Phe	Ile	Pro	Asp	Ile	Lys	Lys	
			210	215						220						
Lys	Gly	Leu	His	Ala	Leu	Pro	Arg	Leu	Val	Leu	Phe	Thr	Ser	Gln	Asp	
225				230						235				240		
Ala	His	Tyr	Ser	Ile	Lys	Lys	Leu	Ser	Ser	Phe	Leu	Gly	Leu	Gly	Thr	
			245				250						255			
Asp	Asn	Val	Tyr	Ala	Ile	Cys	Thr	Asp	Ala	Lys	Gly	Lys	Met	Asp	Val	
			260				265				270					
Gly	His	Leu	Val	Glu	Glu	Ile	Glu	Arg	Ala	Leu	Glu	Glu	Gly	Ala	Ala	
			275	280						285						
Pro	Phe	Met	Val	Ser	Ala	Thr	Ser	Gly	Thr	Thr	Val	Ile	Gly	Ala	Phe	
			290	295						300						
Asp	Pro	Leu	Asp	Glu	Ile	Ala	Asp	Val	Cys	Gln	Lys	Tyr	Gly	Leu	Trp	
305				310						315			320			
Met	His	Val	Asp	Ala	Ala	Trp	Gly	Gly	Gly	Ala	Leu	Met	Ser	Lys	Lys	
			325				330						335			
His	Arg	His	Leu	Leu	Lys	Gly	Val	Glu	Arg	Ala	Asp	Ser	Val	Thr	Trp	
			340				345				350					
Asn	Pro	His	Lys	Leu	Leu	Thr	Ala	Pro	Gln	Gln	Cys	Ser	Thr	Leu	Leu	
			355	360						365						
Leu	Arg	His	Glu	Gly	Leu	Leu	Ala	Glu	Cys	Asn	Ser	Ala	Asn	Ala	Thr	
			370	375						380						
Tyr	Leu	Phe	Gln	Lys	Asp	Lys	Phe	Tyr	Asp	Thr	Lys	Tyr	Asp	Leu	Gly	
385				390						395			400			
Asp	Lys	His	Ile	Gln	Cys	Gly	Arg	Arg	Pro	Asp	Val	Leu	Lys	Phe	Trp	

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420					425					430					
Lys	Val	Phe	Glu	Asn	Thr	Lys	Tyr	Phe	Thr	Asp	Ser	Ile	Lys	Asn	Arg
	435						440					445			
Pro	Gly	Phe	Glu	Leu	Val	Val	Pro	Glu	Pro	Glu	Cys	Thr	Asn	Ile	Cys
	450						455				460				
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465				470					475					480	
Tyr	Lys	Glu	Lys	Leu	His	Lys	Val	Ala	Pro	Lys	Ile	Lys	Glu	Arg	Met
			485						490					495	
Met	Lys	Glu	Gly	Ser	Met	Met	Val	Thr	Tyr	Gln	Pro	Leu	Arg	Glu	Val
		500						505					510		
Pro	Asn	Phe	Phe	Arg	Ile	Val	Phe	Gln	Asn	Ser	Gly	Leu	Asn	Lys	Thr
	515						520					525			
Asp	Met	Thr	His	Leu	Ile	Glu	Glu	Phe	Glu	Arg	Leu	Gly	His	Asp	Leu
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<210> SEQ ID NO 4
 <211> LENGTH: 568
 <212> TYPE: PRT
 <213> ORGANISM: Anopheles arabiensis

<400> SEQUENCE: 4

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		20						25					30		
Glu	Asp	Val	Gln	Gln	Leu	Phe	Val	Ser	Gly	Ala	Asp	Arg	Val	Thr	Ser
		35					40					45			
Val	Leu	Pro	Lys	Lys	Ser	Asp	Ile	Arg	Lys	Ala	Ser	Gln	Val	Asp	Glu
	50					55					60				
Gln	Ala	Ala	Ala	Ala	Ala	Ala	Ala	Ala	Ala	Val	Ser	Glu	Lys	Arg	Tyr
65				70					75					80	
Ala	Ser	Leu	Pro	Asn	Arg	Glu	Gln	His	Gln	Gln	Phe	Leu	Thr	Gln	Phe
			85					90						95	
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		100						105					110		
Asn	Lys	Val	Leu	Asn	Trp	Val	Asp	Pro	Glu	Glu	Leu	Gln	Arg	Thr	Leu
		115					120					125			
Asp	Leu	Ala	Leu	Lys	Asp	Glu	Pro	Asp	Thr	His	Glu	Lys	Leu	Leu	Glu
	130					135					140				
Leu	Thr	Arg	Ala	Thr	Ile	Arg	His	Ser	Val	Lys	Thr	Gly	His	Pro	Tyr
145				150						155				160	
Phe	Met	Asn	Gln	Leu	Phe	Ser	Ser	Val	Asp	Pro	Tyr	Gly	Phe	Ala	Gly
			165						170					175	
Gln	Val	Leu	Thr	Asp	Ala	Leu	Asn	Pro	Ser	Val	Tyr	Thr	Tyr	Glu	Val
		180						185					190		
Ser	Pro	Val	Phe	Val	Leu	Met	Glu	Glu	Val	Val	Leu	Arg	Glu	Met	Arg
		195						200					205		
Thr	Ile	Val	Gly	Tyr	Pro	Asp	Gly	Glu	Gly	Asp	Gly	Ile	Phe	Ala	Pro
	210					215						220			
Gly	Gly	Ser	Met	Ala	Asn	Gly	Tyr	Ala	Ile	Ser	Cys	Ala	Arg	His	Lys
225					230					235					240

-continued

Phe	Met	Pro	Asp	Ile	Lys	Thr	Lys	Gly	Leu	His	Ala	Leu	Pro	Arg	Leu	245	250	255
Val	Ile	Phe	Thr	Ser	Glu	Asp	Ala	His	Tyr	Ser	Val	Lys	Lys	Leu	Ala	260	265	270
Ser	Phe	Met	Gly	Ile	Gly	Ser	Asp	Asn	Val	Tyr	Ala	Ile	Lys	Thr	Asp	275	280	285
Asn	Val	Gly	Lys	Ile	Arg	Val	Glu	His	Leu	Glu	Ser	Glu	Ile	Leu	Arg	290	295	300
Ala	Lys	Ser	Glu	Gly	Ala	Leu	Pro	Phe	Met	Val	Ser	Ala	Thr	Ala	Gly	305	310	315
Thr	Thr	Val	Ile	Gly	Ala	Phe	Asp	Pro	Leu	Glu	Gln	Ile	Ala	Asp	Leu	325	330	335
Cys	Ala	Lys	Tyr	Asn	Leu	Trp	Met	His	Val	Asp	Ala	Ala	Trp	Gly	Gly	340	345	350
Gly	Ala	Leu	Met	Ser	Lys	Lys	Tyr	Arg	Thr	Leu	Leu	Lys	Gly	Ile	Glu	355	360	365
Arg	Ser	Asp	Ser	Val	Thr	Trp	Asn	Pro	His	Lys	Leu	Leu	Ala	Ala	Pro	370	375	380
Gln	Gln	Cys	Ser	Thr	Leu	Leu	Thr	Arg	His	Arg	Asn	Ile	Leu	Ala	Glu	385	390	395
Ala	His	Ser	Thr	Asn	Ala	Thr	Tyr	Leu	Phe	Gln	Lys	Asp	Lys	Phe	Tyr	405	410	415
Asp	Thr	Arg	Tyr	Asp	Thr	Gly	Asp	Lys	His	Ile	Gln	Cys	Gly	Arg	Arg	420	425	430
Ala	Asp	Val	Leu	Lys	Phe	Trp	Phe	Met	Trp	Arg	Ala	Lys	Gly	Thr	Ala	435	440	445
Gly	Phe	Glu	Ala	His	Ile	Asp	Lys	Val	Phe	Glu	Asn	Ala	Glu	His	Phe	450	455	460
Thr	Ser	Ser	Ile	Lys	Ala	Arg	Pro	Gly	Phe	Glu	Met	Val	Ile	Glu	Gln	465	470	475
Pro	Glu	Cys	Thr	Asn	Val	Cys	Phe	Trp	Tyr	Val	Pro	Pro	Gly	Leu	Arg	485	490	495
Gly	Val	Pro	Arg	Asp	Ser	Ala	Glu	Tyr	Arg	Asp	Arg	Leu	His	Lys	Val	500	505	510
Ala	Pro	Lys	Val	Lys	Glu	Arg	Met	Met	Lys	Asp	Gly	Ser	Met	Met	Ile	515	520	525
Thr	Tyr	Gln	Pro	Ile	His	Asp	Lys	Pro	Asn	Phe	Phe	Arg	Leu	Val	Leu	530	535	540
Gln	Asn	Ser	Ser	Leu	Asp	Lys	Ser	Asp	Met	Asn	Tyr	Ile	Ile	Asp	Glu	545	550	555
Ile	Glu	Arg	Leu	Gly	Lys	Asp	Leu									565		

<210> SEQ ID NO 5

<211> LENGTH: 572

<212> TYPE: PRT

<213> ORGANISM: Asbolus verrucosus

<400> SEQUENCE: 5

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Glu	Pro	Ala	Thr	Tyr	Ser	Asp	Ala	Val	Leu	Ser	Ser	Asp	Asp	Glu	Val
			20					25					30		

Cys	Val	Arg	Tyr	Ser	Ser	Gln	Ser	Asp	Thr	Asn	Asn	Ser	Ser	Phe	Tyr
35						40			45						
Gln	Thr	Ala	Thr	Lys	Lys	Leu	Ala	Ser	Phe	Glu	Ser	Leu	Pro	Asn	Arg
50						55			60						
Glu	His	His	Glu	Asp	Phe	Ile	Lys	Lys	Cys	Ala	Glu	Ile	Leu	Ile	Arg
65			70						75			80			
Glu	Ala	Val	Phe	Glu	Gly	Thr	Asn	Arg	Lys	Asn	Arg	Val	Leu	Gln	Trp
			85						90			95			
Asn	Ser	Pro	Glu	Glu	Leu	Gln	Lys	Leu	Met	Asp	Phe	Thr	Leu	Arg	Thr
			100			105						110			
Ser	Pro	Ser	Ser	His	Asp	Glu	Leu	Leu	Asp	Leu	Leu	Arg	Asn	Thr	Val
115						120			125						
Asn	Tyr	Ser	Val	Lys	Thr	Gly	His	Pro	Tyr	Phe	Val	Asn	Gln	Leu	Phe
130						135			140						
Ser	Ser	Leu	Asp	Pro	Tyr	Gly	Leu	Val	Gly	Gln	Trp	Ala	Thr	Asp	Ala
145			150						155			160			
Leu	Asn	Pro	Ser	Val	Tyr	Thr	Tyr	Glu	Val	Ser	Pro	Val	Phe	Thr	Leu
			165						170			175			
Met	Glu	Glu	Val	Val	Leu	Arg	Glu	Met	Arg	Thr	Ile	Val	Gly	Phe	Glu
			180			185						190			
Gly	Gly	Arg	Gly	Asp	Gly	Ile	Phe	Cys	Pro	Gly	Gly	Ser	Ile	Ala	Asn
195						200			205						
Gly	Tyr	Ala	Ile	Ser	Cys	Ala	Arg	His	Lys	Phe	Gln	Pro	Glu	Ile	Lys
210			215						220						
Ala	Thr	Ala	Ser	Val	Thr	Asn	Leu	Leu	Lys	Asn	Ile	Ala	Asn	Ile	Ile
225			230						235			240			
Leu	Leu	Leu	Leu	Gln	Thr	Lys	Gly	Leu	His	Ser	Leu	Pro	Arg	Leu	Val
			245						250			255			
Leu	Phe	Thr	Ser	Glu	Asp	Ala	His	Tyr	Ser	Ile	Lys	Lys	Leu	Ser	Ser
			260			265						270			
Phe	Leu	Gly	Ile	Gly	Thr	Asp	Asn	Val	Tyr	Leu	Ile	Arg	Thr	Asp	Asp
275						280			285						
Arg	Gly	Arg	Met	Asp	Pro	Ser	His	Leu	Ile	Gln	Glu	Ile	Glu	Arg	Ala
290						295			300						
Leu	Ala	Glu	Gly	Gly	Ala	Pro	Phe	Met	Val	Ser	Ala	Thr	Ala	Gly	Thr
305			310						315			320			
Thr	Val	Ile	Gly	Ala	Phe	Asp	Pro	Ile	Asp	Gln	Ile	Ala	Asp	Ile	Cys
			325						330			335			
Glu	Lys	Tyr	Asn	Leu	Trp	Leu	His	Val	Asp	Ala	Ala	Trp	Gly	Gly	Gly
			340			345						350			
Ala	Leu	Met	Ser	Ser	Lys	His	Arg	Ser	Leu	Leu	Lys	Gly	Ile	Glu	Arg
355						360			365						
Ala	Asp	Ser	Val	Thr	Trp	Asn	Pro	His	Lys	Leu	Leu	Thr	Ala	Pro	Gln
370						375			380						
Gln	Cys	Ser	Thr	Leu	Leu	Leu	Arg	His	Glu	Gly	Leu	Leu	Ser	Glu	Thr
385			390						395			400			
His	Ser	Thr	His	Ala	Ala	Tyr	Leu	Phe	Gln	Lys	Asp	Lys	Phe	Tyr	Asp
			405						410			415			
Thr	Lys	Phe	Asp	Thr	Gly	Thr	Lys	Lys	Phe	Asn	Gly	Asp	Lys	His	Ile
			420			425			430						

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Gln	Cys	Gly	Arg	Arg	Ala	Asp	Val	Leu	Lys	Phe	Trp	Phe	Met	Trp	Lys
		435					440					445			
Ala	Lys	Gly	Thr	Leu	Gly	Phe	Glu	Lys	His	Ile	Asn	Lys	Val	Phe	Asp
	450					455					460				
Asn	Ala	Lys	Phe	Phe	Ala	Asp	Ser	Ile	Arg	Asn	Arg	Val	Gly	Phe	Glu
465					470					475					480
Met	Leu	Ile	Asp	Gln	Pro	Glu	Cys	Thr	Asn	Val	Cys	Phe	Trp	Tyr	Ile
			485						490					495	
Pro	Glu	Ser	Leu	Arg	Asn	Ala	Lys	Gln	Asp	Ser	Asp	Tyr	Lys	Glu	Arg
			500					505					510		
Leu	His	Lys	Val	Ala	Pro	Lys	Ile	Lys	Glu	Arg	Met	Met	Lys	Glu	Gly
		515					520					525			
Ser	Met	Met	Val	Thr	Tyr	Gln	Ala	Gln	Lys	Ser	His	Pro	Asn	Phe	Phe
	530					535					540				
Arg	Ile	Val	Phe	Gln	Ser	Ser	Gly	Leu	Asp	Arg	Ala	Asp	Met	Leu	His
545					550					555					560
Leu	Ile	Glu	Glu	Phe	Glu	Arg	Leu	Gly	Arg	Asp	Leu				
				565					570						

<210> SEQ ID NO 6
 <211> LENGTH: 541
 <212> TYPE: PRT
 <213> ORGANISM: Tenebrio molitor

<400> SEQUENCE: 6

Met	Pro	Ala	Arg	Gly	Glu	Gln	Asp	Asp	Val	Val	Gln	Asp	Ile	Ile	Glu
1				5					10					15	
Glu	Pro	Ala	Thr	Tyr	Gly	Asp	Ala	Ile	Leu	Ser	Ser	Asp	Asp	Glu	Val
			20				25						30		
Tyr	Thr	Lys	Phe	Ser	Glu	Arg	Pro	Leu	Thr	Gln	Phe	Tyr	Gln	Pro	Ser
		35					40					45			
Gln	Lys	Arg	Ala	Ser	Phe	Glu	Ser	Leu	Pro	Asn	Arg	Glu	Arg	His	Glu
	50				55						60				
Glu	Phe	Ile	Arg	Lys	Ser	Val	Glu	Ile	Leu	Leu	Lys	Asp	Ala	Val	Phe
65				70					75					80	
Glu	Gly	Thr	Ser	Arg	Asn	Asn	Arg	Val	Leu	Gln	Trp	Thr	Cys	Pro	Glu
			85					90						95	
Glu	Leu	Ser	Arg	Leu	Met	Glu	Phe	Gly	Leu	Lys	Asn	Gly	Pro	Ser	Thr
		100					105						110		
His	Glu	Glu	Leu	Leu	Glu	Ile	Leu	Lys	Lys	Val	Val	Asn	Tyr	Ser	Val
	115					120						125			
Lys	Thr	Gly	His	Pro	Tyr	Phe	Val	Asn	Gln	Leu	Phe	Ser	Ser	Leu	Asp
	130					135					140				
Pro	Tyr	Gly	Leu	Val	Ala	Gln	Trp	Ala	Thr	Asp	Ala	Leu	Asn	Pro	Ser
145					150					155				160	
Val	Tyr	Thr	Tyr	Glu	Val	Ser	Pro	Val	Phe	Ile	Leu	Met	Glu	Glu	Val
			165						170					175	
Val	Leu	Lys	Glu	Met	Arg	Ser	Ile	Val	Gly	Phe	Glu	Ala	Gly	Arg	Gly
			180						185				190		
Asp	Gly	Ile	Phe	Cys	Pro	Gly	Gly	Ser	Ile	Ala	Asn	Gly	Tyr	Ala	Ile
		195					200					205			
Ser	Cys	Ala	Arg	Tyr	Lys	Phe	Gln	Pro	Asp	Ile	Lys	Arg	Lys	Gly	Leu
	210					215					220				

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His Ser Leu Pro Arg Leu Val Leu Phe Thr Ser Glu Asp Ala His Tyr
 225 230 235 240
 Ser Ile Lys Lys Leu Ser Ser Phe Leu Gly Ile Gly Thr Asp Asn Val
 245 250 255
 Tyr Leu Ile Arg Thr Asp Asp Arg Gly Arg Met Asp Val Thr His Leu
 260 265 270
 Ile Gly Gln Ile Glu Arg Ser Leu Ser Glu Gly Ala Ala Pro Phe Met
 275 280 285
 Val Ser Ala Thr Ala Gly Thr Thr Val Ile Gly Ala Phe Asp Pro Leu
 290 295 300
 Asn Glu Ile Ala Ser Val Cys Glu Lys Tyr Lys Leu Trp Leu His Val
 305 310 315 320
 Asp Ala Ala Trp Gly Gly Gly Ala Leu Val Ser Gly Lys His Lys Ser
 325 330 335
 Leu Leu Lys Gly Ile Glu Arg Ala Asp Ser Val Thr Trp Asn Pro His
 340 345 350
 Lys Leu Leu Thr Ala Pro Gln Gln Cys Ser Thr Leu Leu Leu Arg His
 355 360 365
 Glu Gly Ile Leu Ala Ala Ala His Ser Thr Asn Ala Ala Tyr Leu Phe
 370 375 380
 Gln Lys Asp Lys Ser Tyr Asp Thr Lys Phe Asp Thr Gly Asp Lys His
 385 390 395 400
 Ile Gln Cys Gly Arg Arg Ala Asp Val Leu Lys Phe Trp Phe Met Trp
 405 410 415
 Lys Ala Lys Gly Thr Ser Gly Leu Glu Lys His Ile Asn Lys Val Phe
 420 425 430
 Glu Asn Ala Ala Tyr Phe Ala Asp Ser Ile Arg Asn Arg Glu Gly Phe
 435 440 445
 Glu Met Val Ile Asp Gln Pro Glu Cys Thr Asn Val Cys Phe Trp Tyr
 450 455 460
 Ile Pro Glu Ser Leu Arg Ser Cys Lys Gln Asp Ser Asp Tyr Lys Glu
 465 470 475 480
 Arg Leu His Lys Val Ala Pro Lys Ile Lys Glu Arg Met Met Lys Glu
 485 490 495
 Gly Ser Met Met Val Thr Tyr Gln Ala Gln Lys Gln His Pro Asn Phe
 500 505 510
 Phe Arg Ile Val Phe Gln Asn Ser Gly Leu Asp Lys Ala Asp Met Ile
 515 520 525
 His Phe Val Glu Glu Ile Glu Arg Leu Gly Lys Asp Leu
 530 535 540

<210> SEQ ID NO 7

<211> LENGTH: 547

<212> TYPE: PRT

<213> ORGANISM: *Cryptotermes secundus*

<400> SEQUENCE: 7

Met Pro Ala Ser Ser Gly Ile Ile Thr Leu Thr Gln Ser Leu Glu Asn
 1 5 10 15
 Leu Asn Gly Lys His Gly Ile Ser Gly Ser Tyr Glu Asp Met Thr Ala
 20 25 30
 Gly Val Asn Val Ala Val Pro Ser Leu Ser Pro Ser Pro Gly Tyr Val

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35					40					45					
Thr	Glu	Lys	Lys	Ser	Thr	Arg	Ser	Val	Ala	Trp	Phe	Ala	Ser	Leu	Pro
50						55					60				
Asp	Arg	Gln	Arg	His	Ser	Gln	Phe	Leu	Lys	Glu	Ala	Val	Asp	Leu	Met
65					70					75					80
Leu	Asp	Lys	Ala	Val	Phe	Asp	Ala	Ala	Ser	Arg	Thr	Asn	Arg	Val	Val
				85					90					95	
Glu	Trp	Arg	Ser	Pro	Glu	Glu	Leu	Lys	Lys	Leu	Ile	Asp	Leu	Asp	Leu
			100					105					110		
Pro	Ala	Asp	Arg	Val	Ser	His	Asp	Arg	Leu	Leu	Gln	Leu	Leu	Lys	Asp
		115					120					125			
Ile	Ile	Gln	Tyr	Ser	Val	Lys	Thr	Gly	His	Pro	Tyr	Phe	Val	Asn	Gln
	130					135					140				
Leu	Phe	Ser	Ser	Val	Asp	Pro	Tyr	Gly	Leu	Val	Gly	Gln	Trp	Leu	Gly
145					150					155					160
Asp	Ala	Leu	Asn	Pro	Ser	Val	Tyr	Thr	Tyr	Glu	Val	Ser	Pro	Val	Phe
			165						170					175	
Thr	Leu	Met	Glu	Glu	Thr	Val	Leu	Cys	Glu	Met	Arg	Arg	Ile	Val	Gly
		180						185					190		
Phe	Pro	Glu	Gly	Arg	Gly	Asp	Gly	Ile	Phe	Cys	Pro	Gly	Gly	Ser	Ile
	195						200					205			
Ala	Asn	Gly	Tyr	Ala	Ile	Ser	Cys	Ala	Arg	Tyr	Asn	Phe	Val	Pro	Asp
	210					215					220				
Val	Lys	Lys	Arg	Gly	Leu	His	Gly	Leu	Pro	Arg	Leu	Val	Leu	Phe	Thr
225					230					235					240
Ser	Glu	Asp	Ala	His	Tyr	Ser	Ile	Lys	Lys	Met	Ala	Ser	Leu	Leu	Gly
			245						250					255	
Leu	Gly	Ser	Asp	Asn	Val	Tyr	Leu	Ile	His	Cys	Asn	Ser	Lys	Gly	Lys
		260						265					270		
Met	Asp	Val	Gln	His	Leu	Glu	Gln	Glu	Ile	Gln	Arg	Ala	Leu	Glu	Glu
	275						280					285			
Gly	Ala	Ala	Pro	Phe	Met	Val	Ser	Ala	Thr	Ala	Gly	Thr	Thr	Val	Leu
	290					295					300				
Gly	Ala	Phe	Asp	Pro	Ile	Pro	Lys	Ile	Ala	Asp	Ile	Cys	Ser	Lys	Tyr
305					310					315					320
Lys	Met	Trp	Leu	His	Val	Asp	Ala	Ala	Trp	Gly	Gly	Gly	Ala	Leu	Val
			325						330					335	
Ser	Lys	Lys	His	Lys	His	Leu	Leu	Glu	Gly	Ile	Glu	Lys	Ala	Asp	Ser
		340						345					350		
Val	Thr	Trp	Asn	Pro	His	Lys	Leu	Leu	Thr	Ala	Pro	Gln	Gln	Cys	Ser
	355						360					365			
Thr	Phe	Leu	Leu	Arg	His	Glu	Gly	Val	Leu	Ser	Ala	Cys	His	Ser	Ala
	370					375					380				
Ser	Ala	Gln	Tyr	Leu	Phe	Gln	Lys	Asp	Lys	Phe	Tyr	Asp	Thr	Gln	Tyr
385					390					395					400
Asp	Thr	Gly	Asp	Lys	His	Ile	Gln	Cys	Gly	Arg	Arg	Ala	Asp	Val	Leu
			405						410					415	
Lys	Phe	Trp	Phe	Met	Trp	Lys	Ala	Lys	Gly	Thr	Val	Gly	Leu	Glu	Glu
		420						425					430		
His	Ile	Asp	Thr	Val	Phe	Asp	Asn	Ala	Ala	Tyr	Phe	Thr	Lys	Gln	Ile
	435						440					445			

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Lys Lys Arg Glu Gly Phe Arg Met Val Leu Gln Glu Pro Glu Cys Thr
 450 455 460
 Asn Val Cys Phe Trp Tyr Ile Pro Pro Ser Leu Arg Gly His Glu Asp
 465 470 475 480
 Gln Ser Asp Phe Ser Glu Arg Leu His Lys Val Ala Pro Arg Ile Lys
 485 490 495
 Glu Arg Met Ile Lys Glu Gly Ser Met Met Val Thr Tyr Gln Pro Leu
 500 505 510
 Arg Asp Gln Pro Asn Phe Phe Arg Leu Val Leu Gln Asn Ser Gly Leu
 515 520 525
 Asp Trp Ala Asp Met Asp Tyr Phe Val Gln Glu Phe Glu Arg Leu Gly
 530 535 540
 Ser Asp Leu
 545

<210> SEQ ID NO 8
 <211> LENGTH: 575
 <212> TYPE: PRT
 <213> ORGANISM: *Drosophila melanogaster*

<400> SEQUENCE: 8

Met Leu Ala Ser Glu Asn Phe Pro Thr His His Phe Lys Glu Ser Ile
 1 5 10 15
 Phe Lys Pro Tyr Ser Thr Thr Ser Gly Asp Asp Leu Ala Ser Val Ser
 20 25 30
 Pro Leu Thr Ala Thr Ala Ala Leu Val Ala Ser Thr Ser Ser Pro Ala
 35 40 45
 Asp Ser Thr Ser Thr Val Ala Phe Glu Gln Ala Ser Lys Met Leu Ala
 50 55 60
 Asn Ala Ala Asn Asn Asn Asn Asn Asn Asn Asn Ile Thr Ser Thr
 65 70 75 80
 Lys Asp Asp Leu Ser Ser Phe Val Ala Ser His Pro Ala Ala Glu Phe
 85 90 95
 Glu Gly Phe Ile Arg Ala Cys Val Asp Glu Ile Ile Lys Leu Ala Val
 100 105 110
 Phe Gln Gly Thr Asn Arg Ser Ser Lys Val Val Glu Trp His Glu Pro
 115 120 125
 Ala Glu Leu Arg Gln Leu Phe Asp Phe Gln Leu Arg Glu Gln Gly Glu
 130 135 140
 Ser Gln Asp Lys Leu Arg Glu Leu Leu Arg Glu Thr Ile Arg Phe Ser
 145 150 155 160
 Val Lys Thr Gly His Pro Tyr Phe Ile Asn Gln Leu Tyr Ser Gly Val
 165 170 175
 Asp Pro Tyr Ala Leu Val Gly Gln Trp Leu Thr Asp Ala Leu Asn Pro
 180 185 190
 Ser Val Tyr Thr Tyr Glu Val Ala Pro Leu Phe Thr Leu Met Glu Glu
 195 200 205
 Gln Val Leu Ala Glu Met Arg Arg Ile Val Gly Phe Pro Asn Gly Gly
 210 215 220
 Gln Gly Asp Gly Ile Phe Cys Pro Gly Gly Ser Ile Ala Asn Gly Tyr
 225 230 235 240
 Ala Ile Ser Cys Ala Arg Tyr Arg His Ser Pro Glu Ser Lys Lys Asn

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245					250					255					
Gly	Leu	Phe	Asn	Ala	Lys	Pro	Leu	Ile	Ile	Phe	Thr	Ser	Glu	Asp	Ala
			260					265					270		
His	Tyr	Ser	Val	Glu	Lys	Leu	Ala	Met	Phe	Met	Gly	Phe	Gly	Ser	Asp
			275					280					285		
His	Val	Arg	Lys	Ile	Ala	Thr	Asn	Glu	Val	Gly	Lys	Met	Arg	Leu	Ser
			290					295					300		
Asp	Leu	Glu	Lys	Gln	Val	Lys	Leu	Cys	Leu	Glu	Asn	Gly	Trp	Gln	Pro
								310					315		
Leu	Met	Val	Ser	Ala	Thr	Ala	Gly	Thr	Thr	Val	Leu	Gly	Ala	Phe	Asp
								325					330		
Asp	Leu	Ala	Gly	Ile	Ser	Glu	Val	Cys	Lys	Lys	Tyr	Asn	Met	Trp	Met
								340					345		
His	Val	Asp	Ala	Ala	Trp	Gly	Gly	Gly	Ala	Leu	Met	Ser	Lys	Lys	Tyr
								355					360		
Arg	His	Leu	Leu	Asn	Gly	Ile	Glu	Arg	Ala	Asp	Ser	Val	Thr	Trp	Asn
								370					375		
Pro	His	Lys	Leu	Leu	Ala	Ala	Ser	Gln	Gln	Cys	Ser	Thr	Phe	Leu	Thr
								385					390		
Arg	His	Gln	Gln	Val	Leu	Ala	Gln	Cys	His	Ser	Thr	Asn	Ala	Thr	Tyr
								405					410		
Leu	Phe	Gln	Lys	Asp	Lys	Phe	Tyr	Asp	Thr	Ser	Phe	Asp	Thr	Gly	Asp
								420					425		
Lys	His	Ile	Gln	Cys	Gly	Arg	Arg	Ala	Asp	Val	Phe	Lys	Phe	Trp	Phe
								435					440		
Met	Trp	Lys	Ala	Lys	Gly	Thr	Gln	Gly	Leu	Glu	Ala	His	Val	Glu	Lys
								450					455		
Val	Phe	Arg	Met	Ala	Glu	Phe	Phe	Thr	Ala	Lys	Val	Arg	Glu	Arg	Pro
								465					470		
Gly	Phe	Glu	Leu	Val	Leu	Glu	Ser	Pro	Glu	Cys	Thr	Asn	Ile	Ser	Phe
								485					490		
Trp	Tyr	Val	Pro	Pro	Gly	Leu	Arg	Glu	Met	Glu	Arg	Asn	Arg	Glu	Phe
								500					505		
Tyr	Asp	Arg	Leu	His	Lys	Val	Ala	Pro	Lys	Val	Lys	Glu	Gly	Met	Ile
								515					520		
Lys	Lys	Gly	Ser	Met	Met	Ile	Thr	Tyr	Gln	Pro	Leu	Arg	Gln	Leu	Pro
								530					535		
Asn	Phe	Phe	Arg	Leu	Val	Leu	Gln	Asn	Ser	Cys	Leu	Glu	Glu	Ser	Asp
								545					550		
Met	Val	Tyr	Phe	Leu	Asp	Glu	Ile	Glu	Ser	Leu	Ala	Gln	Asn	Leu	
								565					570		

<210> SEQ ID NO 9

<211> LENGTH: 563

<212> TYPE: PRT

<213> ORGANISM: Culex quinquefasciatus

<400> SEQUENCE: 9

Met	Pro	Thr	Asn	Gly	Met	Phe	Asp	Val	Ala	Leu	Gln	Val	Ile	Glu	Asp
1				5					10					15	

Ala	Asn	Leu	Ser	Ser	Gly	Ser	Asp	Ser	Ala	Gly	Val	Ser	Glu	Asp	Glu
					20				25					30	

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Asp	Val	Gln	Leu	Phe	Cys	Thr	Thr	Gly	Asn	Val	Val	Ser	Ser	Lys	Pro
	35						40					45			
Leu	Lys	Lys	Pro	Ser	Leu	Lys	Pro	Val	Thr	Thr	Val	Lys	Asp	Glu	Asp
	50					55					60				
Gln	Asn	Lys	Met	Lys	Thr	Asn	Ala	Lys	Arg	Tyr	Ala	Ser	Leu	Pro	Asn
65					70					75					80
Arg	Glu	Gln	His	Gln	Arg	Phe	Leu	Thr	Asp	Phe	Leu	Ser	Glu	Val	Leu
			85						90					95	
Asn	Asn	Ala	Ile	Phe	Asn	Ala	Thr	Asp	Arg	Ser	Asn	Lys	Val	Leu	Asn
			100					105					110		
Trp	Val	Asp	Pro	Glu	Glu	Leu	Lys	Arg	Ser	Ile	Asp	Leu	Ser	Leu	Lys
	115						120					125			
Ala	Glu	Pro	Asp	Ser	Asp	Glu	Lys	Leu	Leu	Glu	Leu	Ala	Arg	Ala	Thr
	130					135						140			
Ile	Asp	His	Ser	Val	Lys	Thr	Gly	His	Pro	Tyr	Phe	Met	Asn	Gln	Leu
145					150					155					160
Phe	Ser	Ser	Val	Asp	Val	Tyr	Gly	Phe	Ala	Gly	Gln	Cys	Leu	Thr	Asp
			165						170					175	
Ala	Leu	Asn	Pro	Ser	Val	Tyr	Thr	Phe	Glu	Val	Ser	Pro	Val	Phe	Val
		180						185					190		
Leu	Met	Glu	Glu	Val	Val	Leu	Lys	Glu	Met	Arg	Thr	Ile	Val	Gly	Phe
	195						200					205			
Pro	Gly	Gly	Val	Gly	Asp	Gly	Ile	Phe	Cys	Pro	Gly	Gly	Ser	Met	Ala
	210					215					220				
Asn	Gly	Tyr	Ala	Ile	Ser	Cys	Ala	Arg	Phe	Lys	His	Met	Pro	Asp	Val
225					230					235					240
Lys	Thr	Lys	Gly	Leu	His	Ser	Leu	Pro	Arg	Leu	Val	Ile	Phe	Thr	Ser
			245						250					255	
Glu	Asp	Ala	His	Tyr	Ser	Ile	Lys	Lys	Leu	Ala	Ser	Phe	Met	Gly	Ile
		260						265					270		
Gly	Ser	Asp	Asn	Val	Tyr	Pro	Ile	Arg	Thr	Asp	Ala	Val	Gly	Lys	Ile
	275						280					285			
Gln	Pro	Asp	His	Leu	Glu	Ala	Glu	Ile	Leu	Arg	Ala	Lys	Ser	Glu	Gly
	290					295					300				
Ala	Leu	Pro	Phe	Met	Val	Ser	Ala	Thr	Ala	Gly	Thr	Thr	Val	Ile	Gly
305					310					315					320
Ala	Phe	Asp	Pro	Leu	Glu	Gln	Ile	Ala	Asp	Leu	Cys	Gln	Lys	Tyr	Asn
			325						330					335	
Leu	Trp	Met	His	Val	Asp	Ala	Ala	Trp	Gly	Gly	Gly	Ala	Leu	Met	Ser
		340						345					350		
Lys	Lys	Tyr	Arg	Thr	Leu	Leu	Lys	Gly	Val	Glu	Arg	Ala	Asp	Ser	Val
		355					360					365			
Thr	Trp	Asn	Pro	His	Lys	Leu	Leu	Ala	Ala	Pro	Gln	Gln	Cys	Ser	Thr
	370					375					380				
Phe	Leu	Thr	Arg	His	Glu	Gly	Ile	Leu	Ser	Gly	Cys	His	Ser	Thr	Asn
385					390					395					400
Ala	Thr	Tyr	Leu	Phe	Gln	Lys	Asp	Lys	Phe	Tyr	Asp	Thr	Gln	Tyr	Asp
			405						410					415	
Thr	Gly	Asp	Lys	His	Ile	Gln	Cys	Gly	Arg	Arg	Ala	Asp	Val	Leu	Lys
		420						425					430		
Phe	Trp	Phe	Met	Trp	Arg	Ala	Lys	Gly	Thr	Ser	Gly	Phe	Glu	Gln	His

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435	440	445
Ile Asp Lys Val Phe Glu Asn Ala Glu Tyr Phe Thr Asn Ser Ile Lys		
450	455	460
Ala Arg Pro Gly Phe Glu Met Val Ile Glu Asn Pro Glu Cys Thr Asn		
465	470	475 480
Val Cys Phe Trp Tyr Val Pro Pro Gly Leu Arg Gln Val Pro Arg Asp		
	485	490 495
Ser Ala Glu Phe Gly Glu Arg Leu His Lys Val Ala Pro Lys Val Lys		
	500	505 510
Glu Arg Met Met Arg Glu Gly Ser Met Met Ile Thr Tyr Gln Pro Ile		
	515	520 525
His Asp Lys Pro Asn Phe Phe Arg Leu Val Leu Gln Asn Ser Gly Leu		
	530	535 540
Asp Lys Ser Asp Met Asn Tyr Ile Ile Asp Glu Ile Glu Arg Leu Ala		
	545	550 555 560
Ser Asp Leu		
<210> SEQ ID NO 10		
<211> LENGTH: 560		
<212> TYPE: PRT		
<213> ORGANISM: Aedes albopictus		
<400> SEQUENCE: 10		
Met Pro Ala Asn Gly Met Phe Asp Val Ala Leu Gln Val Ile Asp Asp		
1	5	10 15
Ser Asn Val Ser Ser Gly Ser Asp Ser Ala Gly Val Ser Glu Asp Glu		
	20	25 30
Asp Val Gln Leu Phe Cys Ser Met Gly Asn Thr Ile Ala Pro Lys Pro		
	35	40 45
Leu Lys Lys Ser Ile Thr Lys Thr Lys Asp Glu Glu Phe Ser Lys Thr		
	50	55 60
Ala Lys Ala Asn Glu Lys Arg Tyr Ala Ser Leu Pro Asn Arg Glu Gln		
	65	70 75 80
His Gln Gln Phe Leu Thr Asp Phe Leu Ser Glu Val Leu Asn Asn Ala		
	85	90 95
Val Phe Asn Ala Thr Glu Arg Ala Asn Lys Val Leu Asn Trp Val Asp		
	100	105 110
Pro Glu Gln Leu Lys Arg Thr Leu Asp Leu Glu Leu Lys Asp Glu Pro		
	115	120 125
Asp Ser His Glu Lys Leu Leu Glu Leu Thr Arg Ala Thr Ile Lys His		
	130	135 140
Ser Val Lys Thr Gly His Pro Tyr Phe Met Asn Gln Leu Phe Ser Ser		
	145	150 155 160
Val Asp Pro Tyr Gly Phe Ala Gly Gln Ile Leu Thr Asp Ala Leu Asn		
	165	170 175
Pro Ser Val Tyr Thr Phe Glu Val Ser Pro Val Phe Val Leu Met Glu		
	180	185 190
Glu Val Val Leu Lys Glu Met Arg Thr Ile Val Gly Tyr Pro Asp Gly		
	195	200 205
Ala Gly Asp Gly Ile Phe Cys Pro Gly Gly Ser Met Ala Asn Gly Tyr		
	210	215 220
Ser Ile Ser Cys Ala Arg Phe Lys His Met Pro Asp Val Lys Thr Lys		

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225	230	235	240
Gly Leu His Ser Leu Pro Arg Leu Val Ile Phe Thr Ser Glu Asp Ala	245	250	255
His Tyr Ser Val Lys Lys Leu Ala Ser Phe Met Gly Ile Gly Ser Asp	260	265	270
Asn Val Tyr Pro Ile Arg Thr Asp Ala Ile Gly Lys Ile Arg Val Asp	275	280	285
His Leu Glu Ser Glu Ile Leu Arg Ala Lys Ala Glu Gly Ala Val Pro	290	295	300
Phe Met Val Ser Ala Thr Ala Gly Thr Thr Val Ile Gly Ala Phe Asp	305	310	315
Pro Leu Glu Gln Ile Ala Asp Leu Cys Lys Lys Tyr Asn Leu Trp Met	325	330	335
His Val Asp Ala Ala Trp Gly Gly Gly Ala Leu Met Ser Lys Lys Tyr	340	345	350
Arg Ser Leu Leu Lys Gly Ile Glu Arg Ser Asp Ser Val Thr Trp Asn	355	360	365
Pro His Lys Leu Leu Ala Ala Pro Gln Gln Cys Ser Thr Phe Leu Thr	370	375	380
Arg His Glu Gly Ile Leu Ser Glu Cys His Ser Thr Asn Ala Thr Tyr	385	390	395
Leu Phe Gln Lys Asp Lys Phe Tyr Asp Thr Gln Tyr Asp Thr Gly Asp	405	410	415
Lys His Ile Gln Cys Gly Arg Arg Ala Asp Val Leu Lys Phe Trp Phe	420	425	430
Met Trp Arg Ala Lys Gly Thr Ser Gly Leu Glu Gln His Ile Asp Lys	435	440	445
Val Phe Glu Asn Ala Glu His Phe Thr Asn Ser Ile Lys Ala Arg Asp	450	455	460
Gly Phe Glu Met Val Val Glu Thr Pro Glu Cys Thr Asn Val Cys Phe	465	470	475
Trp Tyr Val Pro Pro Gly Leu Arg Ser Val Pro Arg Asp Ser Ala Glu	485	490	495
Phe Thr Glu Arg Leu His Lys Val Ala Pro Lys Val Lys Glu Arg Met	500	505	510
Met Arg Glu Gly Ser Met Met Ile Thr Tyr Gln Pro Ile His Asp Lys	515	520	525
Pro Asn Phe Phe Arg Leu Val Leu Gln Asn Ser Ala Leu Asp Lys Ser	530	535	540
Asp Met Asn Tyr Ile Ile Asp Glu Ile Glu Arg Leu Ala Ala Asp Leu	545	550	555
			560

<210> SEQ ID NO 11

<211> LENGTH: 562

<212> TYPE: PRT

<213> ORGANISM: Aedes aegypti

<400> SEQUENCE: 11

Met Pro Ala Asn Gly Met Phe Asp Val Ala Leu Gln Val Ile Asp Asp	1	5	10	15
Ser Asn Val Ser Ser Gly Ser Asp Ser Ala Gly Val Ser Glu Asp Glu	20	25	30	

Asp	Val	Gln	Leu	Phe	Cys	Ser	Lys	Gly	Asn	Thr	Ile	Val	Pro	Lys	Pro
35							40					45			
Leu	Lys	Lys	Ser	Ile	Ser	Lys	Ile	Lys	Asp	Glu	Glu	Phe	Ser	Lys	Thr
50						55					60				
Ala	Lys	Ala	Asn	Glu	Lys	Arg	Tyr	Ala	Ser	Leu	Pro	Ser	Arg	Glu	His
65					70					75					80
His	Gln	Gln	Phe	Leu	Thr	Asp	Phe	Leu	Ser	Glu	Val	Leu	Asn	Asn	Ala
85									90				95		
Val	Phe	Asn	Ala	Thr	Glu	Arg	Ala	Asn	Lys	Val	Leu	Asn	Trp	Val	Asp
100							105					110			
Pro	Glu	Gln	Leu	Lys	Arg	Thr	Leu	Asp	Leu	Glu	Leu	Lys	Asp	Glu	Pro
115							120					125			
Asp	Ser	His	Glu	Lys	Leu	Leu	Glu	Leu	Thr	Arg	Ala	Thr	Ile	Lys	His
130						135					140				
Ser	Val	Lys	Thr	Gly	His	Pro	Tyr	Phe	Met	Asn	Gln	Leu	Phe	Ser	Ser
145					150					155					160
Val	Asp	Pro	Tyr	Gly	Phe	Ala	Gly	Gln	Ile	Leu	Thr	Asp	Ala	Leu	Asn
165								170						175	
Pro	Ser	Val	Tyr	Thr	Phe	Glu	Val	Ser	Pro	Val	Phe	Val	Leu	Met	Glu
180							185					190			
Glu	Val	Val	Leu	Lys	Glu	Met	Arg	Thr	Ile	Val	Gly	Tyr	Pro	Asp	Gly
195							200				205				
Thr	Gly	Asp	Gly	Ile	Phe	Cys	Pro	Gly	Gly	Ser	Met	Ala	Asn	Gly	Tyr
210						215					220				
Ser	Ile	Ser	Cys	Ala	Arg	Phe	Lys	His	Met	Pro	Asp	Val	Lys	Thr	Lys
225				230						235					240
Gly	Leu	His	Ser	Leu	Pro	Arg	Leu	Val	Ile	Phe	Thr	Ser	Glu	Asp	Ala
245								250					255		
His	Tyr	Ser	Val	Lys	Lys	Leu	Ala	Ser	Phe	Met	Gly	Ile	Gly	Ser	Asp
260							265					270			
Asn	Val	Tyr	Pro	Ile	Arg	Thr	Asp	Ala	Ile	Gly	Lys	Ile	Arg	Val	Asp
275							280					285			
His	Leu	Glu	Ser	Glu	Ile	Leu	Arg	Ala	Lys	Ser	Glu	Gly	Ala	Val	Pro
290						295					300				
Phe	Met	Val	Ser	Ala	Thr	Ala	Gly	Thr	Thr	Val	Ile	Gly	Ala	Phe	Asp
305				310					315						320
Pro	Leu	Glu	Gln	Ile	Ala	Asp	Leu	Cys	Lys	Lys	Tyr	Asn	Leu	Trp	Met
325								330					335		
His	Val	Asp	Ala	Ala	Trp	Gly	Gly	Gly	Ala	Leu	Met	Ser	Lys	Lys	Tyr
340								345				350			
Arg	Ser	Leu	Leu	Lys	Gly	Ile	Glu	Arg	Ser	Asp	Ser	Val	Thr	Trp	Asn
355							360					365			
Pro	His	Lys	Leu	Leu	Ala	Ala	Pro	Gln	Gln	Cys	Ser	Thr	Phe	Leu	Thr
370						375					380				
Arg	His	Glu	Gly	Ile	Leu	Ser	Glu	Cys	His	Ser	Thr	Asn	Ala	Thr</	

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435	440	445
Val Phe Glu Asn Ala Glu His Phe Thr Ser Ser Ile Lys Ala Arg Glu		
450	455	460
Gly Phe Glu Met Val Val Glu Asn Pro Glu Cys Thr Asn Val Cys Phe		
465	470	475 480
Trp Tyr Val Pro Pro Gly Leu Arg Asn Val Pro Arg Asp Ser Ala Glu		
	485	490 495
Phe Thr Glu Arg Leu His Lys Val Ala Pro Lys Val Lys Glu Arg Met		
	500	505 510
Met Arg Glu Gly Ser Met Met Ile Thr Tyr Gln Pro Ile His Asp Lys		
	515	520 525
Pro Asn Phe Phe Arg Leu Val Leu Gln Asn Ser Ala Leu Asp Lys Ser		
	530	535 540
Asp Met Asn Tyr Ile Ile Asp Glu Ile Glu Arg Leu Ala Ala Asp Leu		
545	550	555 560
Lys Pro		
<210> SEQ ID NO 12		
<211> LENGTH: 572		
<212> TYPE: PRT		
<213> ORGANISM: Anopheles sinensis		
<400> SEQUENCE: 12		
Met Pro Ala Asn Gly Val Asn Ser Val Glu Leu Glu Val Ile Glu Asp		
1	5	10 15
Val Ala Thr Thr Tyr Ala Ser Gly Ser Asp Ser Ala Gly Val Ser Glu		
	20	25 30
Asp Glu Asp Val Gln Gln Leu Phe Val Ser Gly Ala His His Ile Ser		
	35	40 45
Ser Val Pro Pro Leu Lys Lys Ala Val Glu Thr Arg Gly Lys Gly Thr		
	50	55 60
Gln Leu Gln Gly Pro Ala Ser Glu Gly Ala Ala Ala Glu Val Ser		
65	70	75 80
Glu Lys Arg Tyr Ala Ser Leu Pro Asn Arg Glu Gln His Gln Gln Phe		
	85	90 95
Leu Thr Asp Phe Leu Thr Glu Val Leu Asn Ser Ala Val Phe Asn Ala		
	100	105 110
Thr Asp Arg Ala Asn Lys Val Leu Asn Trp Val Asp Pro Glu Glu Leu		
	115	120 125
Lys Arg Thr Leu Asp Leu Ala Ile Lys Gln Glu Pro Asp Thr His Glu		
	130	135 140
Lys Leu Leu Glu Leu Thr Arg Ala Thr Ile Arg His Ser Val Lys Thr		
145	150	155 160
Gly His Pro Tyr Phe Met Asn Gln Leu Phe Ser Ser Val Asp Pro Tyr		
	165	170 175
Gly Phe Ala Gly Gln Val Leu Thr Asp Ala Leu Asn Pro Ser Val Tyr		
	180	185 190
Thr Phe Glu Val Ser Pro Val Phe Val Leu Met Glu Glu Val Val Leu		
	195	200 205
Arg Glu Met Arg Thr Ile Val Gly Tyr Pro Asn Gly Glu Gly Asp Gly		
	210	215 220
Ile Phe Ala Pro Gly Gly Ser Met Ala Asn Gly Tyr Ala Ile Ser Cys		

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225	230	235	240
Ala Arg Tyr Lys Phe Met Pro Asp Val Lys Ala Lys Gly Leu His Ala	245	250	255
Leu Pro Arg Leu Val Ile Phe Thr Ser Glu Asp Ala His Tyr Ser Val	260	265	270
Lys Lys Leu Ala Ser Phe Met Gly Ile Gly Ser Asp Asn Val Tyr Ala	275	280	285
Ile Lys Thr Asp Ala Ile Gly Lys Ile Cys Val Asp His Leu Glu Ser	290	295	300
Glu Ile Leu Arg Ala Lys Gln Glu Gly Ala Leu Pro Phe Met Val Ser	305	310	315
Ala Thr Ala Gly Thr Thr Val Ile Gly Ala Phe Asp Pro Leu Glu Gln	325	330	335
Ile Ala Asp Leu Cys Ala Lys Tyr Asn Leu Trp Met His Val Asp Ala	340	345	350
Ala Trp Gly Gly Gly Ala Leu Met Ser Lys Lys Tyr Arg Thr Leu Leu	355	360	365
Lys Gly Ile Glu Arg Ser Asp Ser Val Thr Trp Asn Pro His Lys Leu	370	375	380
Leu Ala Ala Pro Gln Gln Cys Ser Thr Leu Leu Thr Arg His Arg Asn	385	390	395
Ile Leu Ser Glu Cys His Ser Thr Asn Ala Thr Tyr Leu Phe Gln Lys	405	410	415
Asp Lys Phe Tyr Asp Thr Arg Tyr Asp Thr Gly Asp Lys His Ile Gln	420	425	430
Cys Gly Arg Arg Ala Asp Val Leu Lys Phe Trp Phe Met Trp Arg Ala	435	440	445
Lys Gly Thr Ala Gly Phe Glu Gln His Ile Asp Lys Val Phe Glu Asn	450	455	460
Ala Glu His Phe Thr Ser Ser Ile Lys Ala Arg Pro Gly Phe Glu Met	465	470	475
Val Ile Glu Asn Pro Glu Cys Thr Asn Val Cys Phe Trp Tyr Val Pro	485	490	495
Pro Gly Leu Arg Ser Val Pro Arg Asp Ser Ala Glu Phe Arg Glu Arg	500	505	510
Leu His Lys Val Ala Pro Lys Val Lys Glu Arg Met Met Lys Glu Gly	515	520	525
Ser Met Met Ile Thr Tyr Gln Pro Ile His Asp Lys Pro Asn Phe Phe	530	535	540
Arg Leu Val Leu Gln Asn Ser Ser Leu Asp Lys Ser Asp Met Asn Tyr	545	550	555
Ile Ile Asp Glu Ile Glu Arg Leu Gly Lys Asp Leu	565	570	

<210> SEQ ID NO 13

<211> LENGTH: 563

<212> TYPE: PRT

<213> ORGANISM: Anopheles albimanus

<400> SEQUENCE: 13

Met Pro Ala Thr Gly Val Ser Ser Ile Gly Leu Glu Val Gln Glu Glu			
1	5	10	15

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Pro	Ala	Thr	Tyr	Ala	Ser	Gly	Ser	Asp	Ser	Ala	Gly	Val	Ser	Glu	Asp
			20					25					30		
Glu	Asp	Val	Gln	Gln	Leu	Phe	Val	Asn	Gly	Ala	His	Gly	Leu	Thr	Ser
		35					40					45			
Val	Ala	Pro	Ala	Val	Arg	Lys	Pro	Glu	Met	Arg	Gly	Lys	Leu	Ser	Leu
		50				55					60				
Asp	Glu	Ser	Ala	Ala	Ile	Asp	Arg	Lys	Arg	Tyr	Ala	Ser	Leu	Pro	Asn
65					70					75					80
Arg	Glu	Gln	His	Gln	Gln	Phe	Leu	Thr	Glu	Phe	Leu	Thr	Glu	Val	Leu
			85						90					95	
Asn	Ser	Ala	Val	Phe	Asn	Ala	Thr	Asp	Arg	Ala	Asn	Lys	Val	Leu	Asn
			100					105					110		
Trp	Val	Asp	Pro	Glu	Glu	Leu	Ser	Arg	Thr	Leu	Asp	Leu	Ala	Ile	Lys
		115					120					125			
Asp	Glu	Pro	Asp	Thr	His	Glu	Arg	Leu	Leu	Glu	Leu	Thr	Arg	Ala	Thr
		130				135						140			
Ile	Arg	His	Ser	Val	Lys	Thr	Gly	His	Pro	Tyr	Phe	Met	Asn	Gln	Leu
145					150					155					160
Phe	Ser	Ser	Val	Asp	Pro	Tyr	Gly	Phe	Ala	Gly	Gln	Val	Leu	Thr	Asp
			165						170					175	
Ala	Leu	Asn	Pro	Ser	Val	Tyr	Thr	Tyr	Glu	Val	Ser	Pro	Val	Phe	Val
			180					185					190		
Leu	Met	Glu	Glu	Thr	Val	Leu	Arg	Glu	Met	Arg	Lys	Ile	Val	Gly	Tyr
		195					200					205			
Pro	Asn	Gly	Val	Gly	Asp	Ala	Ile	Phe	Ala	Pro	Gly	Gly	Ser	Met	Ala
		210				215					220				
Asn	Gly	Tyr	Ala	Ile	Ser	Cys	Ala	Arg	His	Lys	Phe	Met	Pro	Asp	Ile
225					230					235					240
Lys	Ala	Lys	Gly	Leu	His	Ala	Leu	Pro	Arg	Leu	Val	Ile	Phe	Thr	Ser
			245						250					255	
Glu	Asp	Ala	His	Tyr	Ser	Ile	Lys	Lys	Leu	Ala	Ser	Phe	Met	Gly	Ile
			260					265					270		
Gly	Ser	Asp	Asn	Val	Tyr	Pro	Ile	Lys	Thr	Asp	Glu	Ile	Gly	Lys	Ile
		275					280					285			
Cys	Val	Asp	His	Leu	Glu	Ser	Glu	Ile	Leu	Arg	Ala	Lys	Ala	Glu	Gly
		290				295					300				
Ala	Ser	Pro	Phe	Met	Val	Ser	Ala	Thr	Ala	Gly	Thr	Thr	Val	Ile	Gly
305					310					315					320
Ala	Phe	Asp	Pro	Leu	Glu	Gln	Ile	Ala	Asp	Leu	Cys	Glu	Lys	Tyr	Gln
			325						330					335	
Leu	Trp	Phe	His	Val	Asp	Ala	Ala	Trp	Gly	Gly	Gly	Ala	Leu	Met	Ser
			340					345					350		
Lys	Lys	Tyr	Arg	Thr	Leu	Leu	Lys	Gly	Ile	Glu	Arg	Ser	Asp	Ser	Val
		355					360					365			
Thr	Trp	Asn	Pro	His	Lys	Leu	Leu	Ala	Ala	Pro	Gln	Gln	Cys	Ser	Thr
		370				375					380				
Leu	Leu	Thr	Arg	His	Pro	Asn	Leu	Leu	Ser	Glu	Cys	His	Ser	Thr	Asn
385					390					395					400
Ala	Thr	Tyr	Leu	Phe	Gln	Lys	Asp	Lys	Phe	Tyr	Asp	Thr	Gln	Tyr	Asp
			405						410					415	
Thr	Gly	Asp	Lys	His	Ile	Gln	Cys	Gly	Arg	Arg	Ala	Asp	Val	Leu	Lys

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420	425	430
Phe Trp Phe Met Trp Arg Ala Lys Gly Ser Thr Gly Phe Glu Gln His		
435	440	445
Ile Asp Lys Val Phe Glu Asn Ala Glu Tyr Phe Thr Arg Ser Ile Lys		
450	455	460
Ala Arg Pro Gly Phe Glu Met Val Ile Glu His Pro Glu Cys Thr Asn		
465	470	475
Val Cys Phe Trp Tyr Val Pro Pro Ser Leu Arg Asp Met Ala Arg Asp		
485	490	495
Ser Ala Glu Tyr Arg Glu Arg Leu His Lys Val Ala Pro Lys Val Lys		
500	505	510
Glu Arg Met Met Lys Glu Gly Ser Met Met Ile Thr Tyr Gln Pro Ile		
515	520	525
His Asp Lys Pro Asn Phe Phe Arg Leu Val Leu Gln Asn Ser Ser Leu		
530	535	540
Asp Lys Ser Asp Met Asn Tyr Ile Ile Asp Glu Ile Glu Arg Leu Gly		
545	550	555
		560
Lys Asp Leu		

<210> SEQ ID NO 14

<211> LENGTH: 624

<212> TYPE: PRT

<213> ORGANISM: Anopheles darlingi

<400> SEQUENCE: 14

Met Pro Ala Thr Gly Val Ser Ser Ile Gly Leu Glu Val His Glu Glu		
1	5	10
Pro Ala Thr Tyr Ala Ser Gly Ser Asp Ser Ala Gly Val Ser Glu Asp		
20	25	30
Glu Asp Val Gln Gln Leu Phe Val Asn Gly Ala His Gly Val Thr Arg		
35	40	45
Val Ala Pro Ala Ala Arg Lys Ala Glu Met Arg Gly Lys Leu Ser Leu		
50	55	60
Asp Glu Ser Ala Ala Ile Asp Arg Lys Arg Tyr Ala Ser Leu Pro Asn		
65	70	75
Arg Glu Gln His Gln Gln Phe Leu Thr Glu Phe Leu Thr Glu Val Leu		
85	90	95
Asn Ser Ala Val Phe Asn Ala Thr Asp Arg Ala Asn Lys Val Leu Asn		
100	105	110
Trp Val Asp Pro Glu Glu Leu Ser Arg Thr Leu Asp Leu Ala Ile Lys		
115	120	125
Asp Glu Pro Asp Thr His Glu Arg Leu Leu Glu Leu Thr Arg Ala Thr		
130	135	140
Ile Arg His Ser Val Lys Thr Gly His Pro Tyr Phe Met Asn Gln Leu		
145	150	155
Phe Ser Ser Val Asp Pro Tyr Gly Phe Ala Gly Gln Val Leu Thr Asp		
165	170	175
Ala Leu Asn Pro Ser Val Tyr Thr Tyr Glu Val Ser Pro Val Phe Val		
180	185	190
Leu Met Glu Glu Thr Val Leu Arg Glu Met Arg Lys Ile Val Gly Tyr		
195	200	205
Pro Asn Gly Val Gly Asp Ala Ile Phe Ala Pro Gly Gly Ser Met Ala		

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210	215	220
Asn Gly Tyr Ala Ile Ser Cys Ala Arg His Lys Phe Met Pro Asp Ile		
225	230	235 240
Lys Gly Lys Ser Phe Arg Thr Met His Leu Ile Thr Leu Ile Glu Ser		
	245	250 255
Ala Gly Tyr Gly Met Thr Ile Val Ser Gln His Val Thr Thr Val Val		
	260	265 270
Ala Ala Ile Lys Ile Val His Arg Gln Arg Arg Ser Thr Gly Cys Tyr		
	275	280 285
Thr Arg Ser Trp Leu Ile Glu Thr Ile Gly Asn Gln Ala Ser Ala Lys		
	290	295 300
Gly Leu His Ala Leu Pro Arg Leu Val Ile Phe Thr Ser Glu Asp Ala		
305	310	315 320
His Tyr Ser Ile Lys Lys Leu Ala Ser Phe Met Gly Ile Gly Ser Asp		
	325	330 335
Asn Val Tyr Pro Ile Lys Thr Asp Asp Ile Gly Lys Ile Arg Val Asp		
	340	345 350
His Leu Glu Ser Glu Ile Leu Arg Ala Arg Ala Glu Gly Ala Leu Pro		
	355	360 365
Phe Met Val Ser Ala Thr Ala Gly Thr Thr Val Ile Gly Ala Phe Asp		
	370	375 380
Pro Leu Glu Gln Ile Ala Asp Leu Cys Glu Lys Tyr Gln Leu Trp Phe		
385	390	395 400
His Val Asp Ala Ala Trp Gly Gly Gly Ala Leu Met Ser Lys Lys Tyr		
	405	410 415
Arg Thr Leu Leu Lys Gly Ile Glu Arg Ser Asp Ser Val Thr Trp Asn		
	420	425 430
Pro His Lys Leu Leu Ala Ala Pro Gln Gln Cys Ser Thr Leu Leu Thr		
	435	440 445
Arg His Pro Asn Leu Leu Ser Glu Cys His Ser Thr Asn Ala Thr Tyr		
	450	455 460
Leu Phe Gln Lys Asp Lys Phe Tyr Asp Thr Gln Tyr Asp Thr Gly Asp		
465	470	475 480
Lys His Ile Gln Cys Gly Arg Arg Ala Asp Val Leu Lys Phe Trp Phe		
	485	490 495
Met Trp Arg Ala Lys Gly Ser Thr Gly Phe Glu Gln His Ile Asp Lys		
	500	505 510
Val Phe Glu Asn Ala Glu Tyr Phe Thr Arg Ser Ile Lys Ala Arg Pro		
	515	520 525
Gly Phe Glu Met Val Ile Glu His Pro Glu Cys Thr Asn Val Cys Phe		
	530	535 540
Trp Tyr Val Pro Pro Ser Leu Arg Gly Met Ala Arg Asp Ser Ala Glu		
545	550	555 560
Tyr Arg Glu Arg Leu His Lys Val Ala Pro Lys Val Lys Glu Arg Met		
	565	570 575
Met Lys Glu Gly Ser Met Met Ile Thr Tyr Gln Pro Ile His Asp Lys		
	580	585 590
Pro Asn Phe Phe Arg Leu Val Leu Gln Asn Ser Ser Leu Asp Lys Ser		
	595	600 605
Asp Met Asn Tyr Ile Ile Asp Glu Ile Glu Arg Leu Gly Lys Asp Leu		
610	615	620

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<210> SEQ ID NO 15
<211> LENGTH: 568
<212> TYPE: PRT
<213> ORGANISM: Anopheles stephensi

<400> SEQUENCE: 15

Met Pro Ala Asn Gly Val Cys Ser Val Gly Leu Glu Val Ile Glu Asp
1          5          10          15

Asn Ala Ala Thr Tyr Ala Ser Gly Ser Asp Ser Ala Gly Val Ser Glu
20          25          30

Asp Glu Asp Val Gln Gln Leu Phe Val Asn Gly Ala Asp Arg Val Thr
35          40          45

Ser Val Ser Ser Leu Pro Lys Lys Ser Thr Glu Ala Arg Gly Lys Leu
50          55          60

Ser Gln His Gly Asp Asp Gly Lys Pro Ala Val Ala Glu Lys Arg Tyr
65          70          75          80

Ala Ser Leu Pro Asn Arg Glu Gln His Gln Gln Phe Leu Thr Glu Phe
85          90          95

Leu Thr Glu Val Leu Asn Ser Ala Val Phe Asn Ala Thr Asp Arg Ser
100         105         110

Asn Lys Val Leu Asn Trp Val Asp Pro Glu Glu Leu Lys Arg Thr Leu
115         120         125

Asp Leu Ala Ile Lys Asp Glu Pro Asp Thr His Glu Lys Leu Leu Glu
130         135         140

Leu Thr Arg Ala Thr Ile Arg His Ser Val Lys Thr Gly His Pro Tyr
145         150         155         160

Phe Met Asn Gln Leu Phe Ser Ser Val Asp Pro Tyr Gly Phe Ala Gly
165         170         175

Gln Val Leu Thr Asp Ala Leu Asn Pro Ser Val Tyr Thr Phe Glu Val
180         185         190

Ser Pro Val Phe Val Leu Met Glu Glu Val Val Leu Arg Glu Met Arg
195         200         205

Ser Ile Val Gly Tyr Pro Asn Gly Glu Gly Asp Gly Ile Phe Ala Pro
210         215         220

Gly Gly Ser Met Ala Asn Gly Tyr Ala Ile Ser Cys Ala Arg His Lys
225         230         235         240

Phe Met Pro Asp Ile Lys Thr Lys Gly Leu His Ala Leu Pro Arg Leu
245         250         255

Val Ile Phe Thr Ser Glu Asp Ala His Tyr Ser Val Lys Lys Leu Ala
260         265         270

Ser Phe Met Gly Ile Gly Ser Asp Asn Val Tyr Ala Ile Lys Thr Asp
275         280         285

Ser Ile Gly Lys Ile Arg Ile Glu His Leu Glu Ser Glu Ile Leu Arg
290         295         300

Ala Lys Ala Glu Gly Ala Leu Pro Phe Met Val Ser Ala Thr Ala Gly
305         310         315         320

Thr Thr Val Ile Gly Ala Phe Asp Pro Leu Glu Gln Ile Ala Asp Leu
325         330         335

Cys Ala Lys His Asn Leu Trp Met His Val Asp Ala Ala Trp Gly Gly
340         345         350

Gly Ala Leu Met Ser Lys Lys Tyr Arg Thr Leu Leu Lys Gly Ile Glu

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355	360	365
Arg Ser Asp Ser Val Thr Trp Asn Pro His Lys Leu Leu Ala Ala Pro		
370	375	380
Gln Gln Cys Ser Thr Leu Leu Thr Arg His Arg Asn Ile Leu Ser Glu		
385	390	395
Cys His Ser Thr Asn Ala Thr Tyr Leu Phe Gln Lys Asp Lys Phe Tyr		
405	410	415
Asp Thr Arg Tyr Asp Thr Gly Asp Lys His Ile Gln Cys Gly Arg Arg		
420	425	430
Ala Asp Val Leu Lys Phe Trp Phe Met Trp Arg Ala Lys Gly Thr Ala		
435	440	445
Gly Phe Glu Gln His Ile Asp Lys Val Phe Glu Asn Ala Glu His Phe		
450	455	460
Thr Asn Ser Ile Lys Ala Arg Pro Gly Phe Glu Met Val Ile Glu Asn		
465	470	475
Pro Glu Cys Thr Asn Val Cys Phe Trp Tyr Val Pro Pro Gly Leu Arg		
485	490	495
Ser Val Pro Arg Asp Ser Ser Glu Phe Arg Glu Arg Leu His Lys Val		
500	505	510
Ala Pro Lys Val Lys Glu His Met Met Lys Glu Gly Ser Met Met Ile		
515	520	525
Thr Tyr Gln Pro Ile His Asp Lys Pro Asn Phe Phe Arg Leu Val Leu		
530	535	540
Gln Asn Ser Ser Leu Asp Lys Ser Asp Met Asn Tyr Ile Ile Asp Glu		
545	550	555
Ile Glu Arg Leu Gly Lys Asp Leu		
565		

<210> SEQ ID NO 16

<211> LENGTH: 1623

<212> TYPE: DNA

<213> ORGANISM: Tribolium castaneum

<400> SEQUENCE: 16

atgccggcgga ccggcgagga ccaggatctg gttcaagacc tgatcgagga accggcgacc	60
ttcagcgatg cgggtgctgag cagcgacgag gaactgtttc accagaaatg cccgaagccg	120
gcgccgattt acagcccggg gagcaaaccg gttagcttcg aaagcctgcc gaaccgtcgt	180
ctgcacgagg aatttctgcg tagcagcgtg gatgttctgc tgcaggaagc ggtgttcgaa	240
ggcaccaacc gtaagaaccg tgttctgcaa tggcgtgagc cggaggaact gcgtcgtctg	300
atggactttg gtgttcgtag cgcgccgagc acccatgagg aactgctgga agtgctgaag	360
aaagtgggta cctacagcgt gaaaaccggc caccggtatt tcgttaacca gctgttttagc	420
gcggtggatc cgtacgggtc ggttgcgcag tgggcgaccg atgcgctgaa cccgagcgtg	480
tacacctatg aggttagccc ggtgttcggt ctgatggagg aagtgggtct gcgtgagatg	540
cgtgcgatcg ttggtttcga aggtggcaag gccgatggta tcttctgccc ggggtggcagc	600
attgcgaacg gttacgcgat tagctgcgcg cgttatcggt ttatgccgga catcaagaaa	660
aagggtctgc acagcctgcc gcgtctgggt ctgttcacca gcgaagatgc gcactacagc	720
attaaaaagc tggcgagctt tcagggcacg ggtaccgaca acgtgtatct gattcgtacc	780
gatgcgctg gccgtatgga cgtgagccac ctggttgagg aaattgagcg tagcctgcgt	840

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gaggggtgcgg cgccgtttat ggtgagcgcg accgcgggta ccaccgttat tggcgcttt	900
gatccgatcg agaaaattgc ggacgtgtgc caaaaataca agctgtggct gcatgttgat	960
gcggcggtggg gtggcggtgc gctggtgagc gcgaacacc gtcacctgct gaagggcatc	1020
gaacgtgcgg acagcgttac ctggaacccg cacaagctgc tgaccgcgcc gcagcaatgc	1080
agcaccctgc tgctgcgtca cgaggggtgtg ctggcggaag cgcacagcac caacgcggcg	1140
tacctgttcc agaaggataa gttttacgac accaaatatg acaccggcga taagcacatc	1200
caatgcggtc gtcgtgcgga tgttctgaaa ttctggttta tgtggaaagc gaagggcacc	1260
agcggctctgg agaaacacgt ggataagggt ttcgaaaacg cgcgtttctt taccgactgc	1320
attaagaacc gtgagggctt cgaatagggt atcgcgagc cggaatacac caacatttgc	1380
ttttggtacg ttccgaaaag cctgcgtggt cgtaggacg aggcggatta caaagacaag	1440
ctgcacaaaag tggcgccgcg tatcaaagag cgtatgatga aggaaggcag catgatggtt	1500
acctatcagg cgcaaaaagg tcaccgaac ttctttcgtt tcgtgttcca gaacagcggc	1560
ctggataagg cggacatggt gcatctggtg gaggaattg aacgtctggg tagcgacctg	1620
taa	1623

<210> SEQ ID NO 17

<211> LENGTH: 1686

<212> TYPE: DNA

<213> ORGANISM: Culex tarsalis

<400> SEQUENCE: 17

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agcggcagcg acagcgcggg tgtgagcgag gacgaagatg ttcaactgtt cagcaccacc	120
ggtaacatcg tgagcagcaa accgctgaag aaaccggcgc tgaagccggc gaccaaaagc	180
gaagatcaga acaagaccaa agcgaacgcg aagcgttacg cgagcctgcc gaacctgag	240
cagcaccaac gtttctgac cgactttctg agcgaagtgc tgaacggcgc gatctttaac	300
gcgaccgacc gttagcaaaa agtgctgaac tgggttgatc cggagggaact gaagcgtagc	360
attgacctga gcctgaaaga tgagccggac agcgtatgaga agctgctgga actggcgcgt	420
gcgaccatcg accacagcgt gaagaccggt caccctgact tcatgaacca gctgtttagc	480
agcgtggacc cgtatggctt cgcgggtcaa gttctgaccg atgcgctgaa cccgagcgtg	540
tacaccttcg aagttagccc ggtggttgtt ctgatggagg aagtggttct gaaagaaatg	600
cgtaccattg tgggtttccc gggtggcgtt ggcgacggta tcttttgccc gggtggcagc	660
atggcgaaac gctatgcgat tagctgcgcg cgttttaagc acatgccgga cgtgaagacc	720
aaaggtctgc acagcctgcc gcgtctggtt attttcacca gcgaagatgc gcactacagc	780
atcaagaaac tggcgagctt tatgggcacg ggtagcgata acgtgtatcc gattcgtacc	840
gacgcgggtg gcaaaatcca gccggatcac ctggaggcgg aaattctgcg tgcgaagagc	900
gaggggtgcgg tgccgttcat ggttagcgcg accgcgggca ccaccgtgat tggcgcttt	960
gaccgcgtgg aacagatcgc gcatctgtgc caaaaataca acctgtggat gcatgttgat	1020
gcggcggtgg gtggcggtgc gctgatgagc aagaaatata gtacctgct gaaaggtgtg	1080
gagcgtgcgg atagcgttac ctggaacccg cacaagctgc tggcggcgcc gcagcaatgc	1140
agcaccttcc tgaccctgca cgaaggcatt ctgagcgggt gccacagcac caacgcgacc	1200

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tacctgttcc agaaggacaa attttacgat acccaatatg acaccggcga taagcacatt	1260
cagtgcggtc gtcgtgcgga cgttctgaaa ttctgggtta tgtggcgtgc gaagggtacc	1320
agcggctctgg agcaacacat cgataaagtg ttcgagaccg cggaatactt taccaacagc	1380
attaaggcgc gtcggggett cgaaatggtt atcgagaacc cggaatgcac caacgtgtgc	1440
ttttggtatg ttccgccggg tctgcgtcaa gtgccgcgtg acagcgcgga gttcggtgaa	1500
cgtctgcaca aagtgggcc gaagggtaaa gagcgtatga tgcgtgaagg tagcatgatg	1560
atcacctacc agccgattca cgataaacg aacttcttct gtctggttct gcaaacacagc	1620
gcgctggaca agagcgatat gaactatatc attgacgaga tcgaacgtct ggcgagcgat	1680
ctgtaa	1686

<210> SEQ ID NO 18

<211> LENGTH: 1635

<212> TYPE: DNA

<213> ORGANISM: Aethina tumida

<400> SEQUENCE: 18

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agcgatgcgg tggcgagcag cagcgatgat gaaaccgttc aatatagcaa cgacgagcgt	120
agcatccgtg atatgaaagc gaccattgcy accggcaagc tggcgacctt cgaaagcctg	180
ccgagccgtg cgcaccacga ggaatttatc cgtagctgca tggacgtgat tctgaaagag	240
gcgggttttcg atggcaccaa ccgtaacaac ccggtgctga actttgttaa ccggagggaa	300
ctgcaaaagca aagtgaactt caaactgaag accgcgccga gcacccacga agacctgctg	360
aaaaccctga aggataccat tcgttacagc gtgaagaccg gtcacccgta ttctgttaac	420
cagctgttta gcagcctgga ccggtacggc ctgggtgggc aatggctgac cgatgcgctg	480
aaccgcacgg tttaacaccta tgagggtgtc ccggttttta ccctgatgga ggaagaggtg	540
ctgcgtgaaa tgcgtacat cgttggttc aagaacgggtg aaggatgatg tatcttctgc	600
ccgggtggca gcatggcgaa cggttatgcy atcagctgcy cgcgtcacaa attcatccc	660
gatattaaga aaaagggcct gcatgcgctg ccgctctgcy tgctgtttac cagccaggac	720
gcgcactaca gcatcaaaaa gctgagcagc ttctgggtc tgggcaccga taacgtttat	780
gcgatttgca ccgacgcgaa aggcgaagat gatgtgggcc acctggttga agagattgaa	840
cgtgcgctgg aagagggtgc ggcgcggtt atggttagcy cgaccagcgg taccaccgtt	900
atcggcgcgt tcgacccgct ggatgagatt gcggacgtgt gccaaaaata cggctctgtg	960
atgcatgttg atgcggcgtg ggggtggcgt gcgctgatga gcaaaaagca ccgtcacctg	1020
ctgaagggcg tggaaactgc ggacagcgtt acctggaacc cgcacaaact gctgaccgcg	1080
ccgcagcaat gcagaccct gctgctgctg cacgaaggtc tgctggcgga gtgcaacagc	1140
gcgaacgcga cctacctgtt ccagaaagac aagttttacg ataccaaata tgacctgggt	1200
gataagcaca tccaatgcgg ccgtcgtccg gatgtgctga agttctggtt tatgtggaaa	1260
gcgaagggtg ccagcggtt cgaacagcac atcgacaaag ttttcgagaa caccaagtat	1320
tttaccgata gcattaaaaa ccgtccgggt ttgaaactg tggttccgga accggagtgc	1380
accaacattt gcttctggta cgttccgcg agcctgcgtc aagcgaaaag cgaccggat	1440
tataaagaga agctgcacaa ggtggcgccg aaaatcaagg aacgtatgat gaaagagggc	1500

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agcatgatgg ttacctacca gccgctgcgt gaagtgccga acttctttcg tatcgttttt 1560
caaacacagcg gtctgaacaa gaccgacatg acccacctga ttgaagagtt cgagcgtctg 1620
ggccacgatc tgtaa 1635

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<210> SEQ ID NO 19
<211> LENGTH: 1707
<212> TYPE: DNA
<213> ORGANISM: Anopheles arabiensis

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<400> SEQUENCE: 19

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gcgagcggta gcatagcgc gggcgtagc gaggacgaag atgtgcagca actgttcgtt 120
agcgtgtcgg accgtgtgac cagcgttctg ccgaagaaaa gcgacatccg taaagcgagc 180
caggtggatg agcaagcggc ggcggcggcg gcggcggcgg cgggtgtctga gaagcgttat 240
gcgagcctgc cgaaccgtga acagcaccag caattcctga cccaatttct gaccgaagtg 300
ctgaacacgcg cggtttttaa cgcgacgcac cgtgcgaaca aagtgtcga ctgggttgac 360
ccggaggaaac tgcaacgtac cctggatctg gcgctgaagg acgagccgga taccacgag 420
aaactgctgg aactgaccgc tgcgaccatt cgtcacagcg tgaagaccgg tcaccgtac 480
ttcatgaacc agctgtttag cagcgtggac ccgtatggtt tcgcgggcca agttctgacc 540
gatgcgctga acccgagcgt gtacacctat gaagttagcc cgggtgttgt tctgatggag 600
gaagtggctc tgcgtgagat gcgtaccatt gttggctacc cggacggcga aggtgatggt 660
atttttgcgc cgggtggcag catggcgaac ggttatgcga ttagctgcgc gcgtcacaaa 720
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accaccgtta ttggcgcgtt tgaccgcgtg gaacagatcg cggatctgtg cgcgaagtac 1020
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cgtaccctgc tgaagggtat tgagcgtagc gatagcgtta cctggaaccc gcacaaactg 1140
ctggcggcgc cgcagcaatg cagcaccctg ctgaccctgc accgtaacat cctggcggaa 1200
gcgcacagca ccaacgcgac ctacctgttc cagaaggaca aattttacga tacccgttat 1260
gacaccggtg ataagcacat tcaatgcggc cgtcgtgcgg acgttctgaa gttctggttt 1320
atgtggcgtg cgaagggtac cgcgggcttc gaggcgacac tcgataaggt gttcgagaac 1380
gcggaacact ttaccagcag cattaaagcg cgtccgggtt tcgaaatggt tatcgagcaa 1440
ccggaatgca ccaacgtgtg cttttgggtat gttccgccgg gtctgcgtgg cgtgccgcgt 1500
gacagcgcgg agtatcgtga tcgtctgcac aagggtggcg cgaagggtaa agaactgatg 1560
atgaaagacg gtagcatgat gatcacctac cagccgattc acgataagcc gaacttcttt 1620
cgtctgggtc tgcaaaacag cagcctggac aaaagcgata tgaactatat cattgacgag 1680
atcgaacgtc tgggcaagga tctgtaa 1707

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<210> SEQ ID NO 20
<211> LENGTH: 1738

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<212> TYPE: DNA

<213> ORGANISM: *Asbolus verrucosus*

<400> SEQUENCE: 20

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ggcgacctac agcgcgcgcg ttctgagcag cgacgatgaa gtgtgcgttc gttacagcag      120
ccagagcgat accaacaaca gcagctttta tcaaaccgcg accaagaaac tggcgagcct      180
cgagagcctg ccgaaccgtg agcaccacga agactttatc aagaaatgcg cggaaatcct      240
gattcgtgag cgggtgttcg aaggcaccaa ccgtaaaaac cgtgttctgc aatggaacag      300
cccgaggagaa ctgcaaaagc tgatggattt caccctgcgt accagcccga gcagccatga      360
cgaactgctg gatctgtgct gtaacaccgt gaactacagc gttaaaaccg gtcacccgta      420
tttcgtgaac cagctgttca gcagcctgga cccgtacggt ctggtgggtc agtgggagac      480
cgatgcgctg aaccgcgagc ttacaccta tgaggtgtct ccggttttta ccctgatgga      540
ggaagtgtt ctgcgtgaga tgcgtacatc tgtgggcttt gaaggtggcc gtggcgatgg      600
tatcttctgc ccgggtggca gcattgcgaa cggttatgcg atcagctgcg cgcgtcacia      660
gttccagccg gaaattaaag cgaccgcgag cgttaccac ctgctgaaaa acatcgcgaa      720
catcattctg ctgctgtgct aaaccaagg tctgcacagc ctgccgcgctc tgggtgctgt      780
taccagcgag gacgcgcact acagcatcaa gaaactgagc agcttctctg gcattggtac      840
cgataacgtt tatctgatcc gtaccgacga tcgtggtcgt atggaccgga gccacctgat      900
ccaggagatt gagcgtgcgc tggcggaggg tggcgcgcgc tttatggta gcgcgaccgc      960
gggcaccacc gttattggtg cgttcgaccc gatcgatcaa attgcggata tctgcgaaaa      1020
atacaacctg tggctgcatg tggatgcggc gtgggggtggc ggtgcgctga tgagcagcaa      1080
gcaccgtagc ctgctgaaag gcacgcagcg tgcggacagc gttacctgga acccgacaaa      1140
gctgctgacc gcgcgcgagc aatgcagcac cctgctgctg cgtcacgagg gtctgctgag      1200
cgaaaccac agcaccacgc cgcgctacct gttccagaag gacaaatctt atgataccaa      1260
gttcgacacc ggcaccaaga aattcaacgc tgataaacac attcaatgcg gccgtcgtgc      1320
ggacgtgctg aagttctggt ttatgtggaa ggcgaaaggc accctgggtt ttgaaaagca      1380
catcaacaaa gttttcgata acgcgaaatt ctttgcggac agcattcgta accgtgtggg      1440
ttttgagatg ctgatcgatc agccggaatg caccaacgtt tgcttctggt acattccgga      1500
gagcctgcgt aacgcgaagc aagacagcga ttataaggaa cgtctgcaca aagttgcgcc      1560
gaagatcaaa gagcgtatga tgaaagaagg tagcatgatg gtgacctatc aggcgcaaaa      1620
gagccaccgc aacttcttct gtattgtttt tcagagcagc ggctgggacc gtgcggatat      1680
gctgcacctg atcgaggagt tcgagcgtct gggtcgtgac ctgtaatgat aagaattc      1738

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<210> SEQ ID NO 21

<211> LENGTH: 1626

<212> TYPE: DNA

<213> ORGANISM: *Tenebrio molitor*

<400> SEQUENCE: 21

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atgccggcgc gtggcgagca ggatgatgtg gttcaagaca tcattgagga accggcgacc      60
tacggtgatg cgatcctgag cagcgacgat gaggtgtata ccaagttcag cgaacgtccg      120
ctgacccaat tttaccagcc gagccaaaaa cgtgcgagct tcgagagcct gccgaaccgt      180

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gaacgtcacg aggaatttat ccgtaagagc gtggagattc tgctgaaaga cgcggttttc	240
gaaggcacca gccgtaacaa ccgtgttctg caatggacct gcccgaggga actgagccgt	300
ctgatggagt ttggtctgaa gaacggcccc agcaccacac aggaactgct ggaaatcctg	360
aagaaagtgg ttaactacag cgtgaaaacc ggcaccccg atttcgttaa ccagctgttt	420
agcagcctgg acccgatatg tctggttgcg caatgggcga ccgatgcgct gaacccgagc	480
gtgtacacct atgagggtgc tccggttttc attctgatgg aggaagtggg tctgaaggag	540
atgcgtagca tcgtgggttt cgaagcgggc cgtggtgatg gcattctctg cccgggtggc	600
agcattgcga acggttacgc gattagctgc gcgcgttata aattccagcc ggacatcaag	660
cgtaaaggtc tgcacagcct gccgcgtctg gttctgttca ccagcgaaga tgcgcactat	720
agcattaaga aactgagcag ctttctgggt atcggcaccg acaacgttta cctgattcgt	780
accgacgacg gtggtcgtat ggatgtgacc caccctgatcg gccaaattga acgtagcctg	840
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tttgaccgcg tgaacgagat tgcgagcgtg tgcgaaaagt acaaactgtg gctgcatgtt	960
gatgcggcgt ggggtggcgg tgcgctggtt agcggcaagc acaaaagcct gctgaagggc	1020
atcgagcgtg cggacagcgt gacctggaac ccgcacaaac tgctgaccgc gccgcagcaa	1080
tgcagcaccg tgctgctgcg tcacgaaggt attctggctg cggcgcacag caccaacgcg	1140
gcgtacctgt tccagaagga caaaagctat gataccaagt ttgacaccgg tgataaacac	1200
atccaatgcg gccgtcgtgc ggatgtgctg aagttctggt ttatgtggaa ggcgaaaggt	1260
accagcggcc tggagaagca cattaacaaa gttttcgaaa acgcggcgta ttttgcggac	1320
agcatccgta accgtgaggg ctctgaaatg gtgattgacg agccggagtg caccaacggt	1380
tgcttttggt atatcccgga aagcctgcgt agctgcaaac aagacagcga ttacaaggag	1440
cgtctgcaca aagttgcgcc gaagattaaa gagcgtatga tgaagggaagg tagcatgatg	1500
gttacctacc aggcgcacaaa acagcaccgc aacttcttct gtatcgtgtt ccagaacagc	1560
ggcctggaca aggcggatat gatccacttt gttgaggaaa ttgaacgtct gggtaaaagc	1620
ctgtaa	1626

<210> SEQ ID NO 22

<211> LENGTH: 1644

<212> TYPE: DNA

<213> ORGANISM: *Cryptotermes secundus*

<400> SEQUENCE: 22

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cacggcatca gcggtagcta cgaagacatg accgcgggcg tgaacgtggc ggtgccgagc	120
ctgagcccca gcccggttta tgttaccgag aagaaaagca cccgtagcgt ggcgtgggtt	180
gcgagcctgc cggaccgtca gcgtcacagc caatttctga aagaggcggg tgacctgatg	240
ctggataaag cgggtgttca tgcggcgagc cgtaccaacc gtgtggttga atggcgtagc	300
ccggaggaa c tgaagaaact gattgacctg gatctgccgg cggaccgtgt gagccacgat	360
cgtctgctgc aactgctgaa ggacatcatt caatacagcg ttaaaacogg ccaccggtat	420
ttcgtgaacc agctgtttag cagcgttgac ccgtacggcc tgggtgggtca atggctgggt	480
gatgcgctga acccgagcgt ttacacctat gaggtttctc cgggtgtttac cctgatggag	540

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gaaaccgttc tgtgogagat gcgtcgtatt gtgggcttcc cggaaggccg tggatgatgc 600
atcttttgcc cgggtggcag cattgcgaac ggtaacgca tcagctgcgc gcgttataac 660
ttcgttcccg acgtgaagaa acgtggcctg cacggtctgc cgcgtctggt gctgtttacc 720
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gcggcgctact ttaccaagca gatcaagaaa cgtgaaggct tccgtatggt tctgcaagag 1380
ccggaatgca ccaacgtgtg cttttggtat atcccgccga gcctgcgtgg tcacgaggac 1440
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aaagaaggta gcatgatggt tacctaccag ccgctgcgtg accaaccgaa cttctttcgt 1560
ctggtgctgc aaaacagcgg cctggattgg gcggacatgg attatttcgt tcaagagttt 1620
gaacgtctgg gtacgatct gtaa 1644

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<210> SEQ ID NO 23
<211> LENGTH: 1728
<212> TYPE: DNA
<213> ORGANISM: Drosophila melanogaster
<400> SEQUENCE: 23

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gttgcgagca ccagcagccc ggcggatagc accagcaccg tggcgtttga acaggcgagc 180
aagatgctgg cgaacgcggc gaacaataat aacaacaaca acaacaacat caccagcacc 240
aaagacgacg tgagcagctt tgttgcgagc caccggcgcg cggagtttga aggtttcatt 300
cgtgcgtgcg tggacgagat cattaagctg gcggttttcc aaggtaccaa ccgtagcagc 360
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gaacaaggcg agagccagga taagctgcgt gaactgctgc gtgagaccat ccgttttagc 480
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gcgaagccgc tgatcatttt caccagcgaa gacgcgcact acagcgtgga gaaactggcg 840
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atcagcgaag tgtgcaagaa atataacatg tggatgcatg ttgatgcggc gtggggtggc    1080
ggtgcgctga tgagcaagaa ataccgtcac ctgctgaacg gtattgagcg tgcggacagc    1140
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cgtcaccagc aagttctggc gcaatgccac agcaccaacg cgacctacct gttccagaag    1260
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cacgtggaga aggttttctg tatggcggag ttcttcaccg cgaaagtgcg tgaacgtccg    1440
ggctttgagc tggttctgga aagcccgagc tgcaccaaca ttagcttctg gtatgttccg    1500
ccgggtctgc gtgagatgga acgtaaccgt gagttctacg accgtctgca caaagtggcg    1560
ccgaagggta aagagggcat gatcaagaaa ggtagcatga tgattaccta tcaaccgctg    1620
cgtcagctgc cgaacttctt tctctgtgtg ctgcaaaaac gctgcctgga ggaaagcgac    1680
atggtttact tcttgatga gattgaaagc ctggcgacga acctgtaa    1728

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<210> SEQ ID NO 24

<211> LENGTH: 1692

<212> TYPE: DNA

<213> ORGANISM: Culex quinquefasciatus

<400> SEQUENCE: 24

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atgccgacca acggcatggt cgacgtggcg ctgcaagtta ttgaggatgc gaacctgagc    60
agcggcagcg acagcgcggg tgtgagcgag gacgaagatg ttcaactggt ttgcaccacc    120
ggtaacgtgg ttagcagcaa accgctgaag aaaccgagcc tgaagccggt gaccaccgtt    180
aaagacgaag atcagaacaa gatgaaaacc aacgcgaagc gttacgcgag cctgcggaac    240
cgtgagcagc accaacgttt cctgaccgac tttctgagcg aagtgtgaa caacgcgatc    300
ttcaacgcga cgcacgtag caacaaagtg ctgaactggg ttgaccgga ggaactgaag    360
cgtagcattg atctgagcct gaaagcggag ccggacagcg atgagaagct gctggaactg    420
gcgcgtgcga ccatcgacca cagcgtaag accggccacc cgtacttcat gaaccagctg    480
tttagcagcg tggacgttta tggcttcgcg ggtcaatgcc tgaccgatgc gctgaaccgc    540
agcgtgtaca ccttcgaagt tagcccggtg tttgttctga tggagggaagt ggttctgaaa    600
gaaatgcgta ccattgtggg tttcccggtt ggcgttggcg acggtatctt ttgccgggtt    660
ggcagcatgg cgaacggcta tgcgattagc tgcgcgcgtt ttaagcacat gccggacgtg    720
aagaccaaa gttctgcacag cctgcgcgtt ctggttattt tcaccagcga agatgcgcac    780
tacagcatca agaaactggc gagctttatg ggcacggta gcgataacgt gtatccgatt    840
cgtaccgacg cgggtggcaa aatccagccg gatcacctgg aggcggaaa tctgcgtgcg    900
aagagcgagg gtgcgctgcc gtttatggtt agcgcgaccg cgggcaccac cgttattggt    960
gcgtttgacc cgctggaaca gatcgcggat ctgtgccaaa aatacaacct gtggatgcat    1020
gtggatgcgg cgtggggtgg cggtcgctg atgagcaaga aatatcgta cctgctgaaa    1080
ggtgtggagc gtgcggatag cgttacctgg aaccgcaca agctgctggc ggcgcgcgag    1140
caatgcagca ccttctgac ccgtcacgaa ggcattctga gcggttgcca cagcaccaac    1200

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gcgacctacc tgttcagaa ggacaaattt tacgataccc aatatgacac cggcgataag 1260
cacattcagt gcggtcgteg tgcggacgtt ctgaaattct ggtttatgtg gcgtgcgaag 1320
ggcaccagcg gtttcagca acacatcgat aaagtgttcg agaaccgga atactttacc 1380
aacagcatta aggcgcgtcc gggtttcgaa atggttatcg agaaccgga atgcaccaac 1440
gtgtgctttt ggtatgttcc gccgggtctg cgtcaagtgc cgctgacag cgcggagttt 1500
ggtgaacgtc tgcacaaagt ggcgcgaag gttaagagc gtatgatgcg tgaaggcagc 1560
atgatgatca cctaccagcc gattcacgat aaaccgaact tctttcgtct ggttctgcaa 1620
aacagcggtc tggacaagag cgatatgaac tatatcattg acgagatcga acgtctggcg 1680
agcgatctgt aa 1692

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<210> SEQ ID NO 25

<211> LENGTH: 1683

<212> TYPE: DNA

<213> ORGANISM: Aedes albopictus

<400> SEQUENCE: 25

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atgccggcga acggcatgtt cgatgtggcg ctgcaagtta tcgacgatag caacgtgagc 60
agcggtagcg acagcgcggg cgtgagcgag gatgaagatg ttcaactgtt ttgctcgatg 120
ggtaacacca tcgcgccgaa accgtgaag aaaagcatta ccaagaccaa agatgaggaa 180
tttagcaaga ccgcgaaagc gaacgagaag cgttacgcga gcctgccgaa ccgtgaacag 240
caccagcaat tcctgaccga ctttctgagc gaggtgctga acaacgcggg tttcaacgcg 300
accgaacgtg cgaacaaagt gctgaactgg gttgatccgg agcaactgaa gcgtaccctg 360
gacctggagc tgaagacga accggatagc cacgagaagc tgctggaact gacctgtgcg 420
accatcaagc acagcgtgaa aaccggtcac ccgtacttca tgaaccagct gtttagcagc 480
gttgatccgt atggttttgc gggccaaatt ctgaccgacg cgctgaaccc gagcgtgtac 540
accttcgaag ttagcccggt gtttgttctg atggaggaag tggttctgaa agaatgcgt 600
accattgtgg gttaccgga tggtgccggg gatggcattt tctgcccggg tggcagcatg 660
gcgaacgggt atagcatcag ctgcgcgct tttaagcaca tgccggatgt taagaccaa 720
ggcctgcaca gcctgccgcg tctggtgatt ttcaccagcg aggaacgcga ctacagcgtt 780
aagaaactgg cgagctttat gggtatcggc agcgacaacg tgatccgat tcgtaccgat 840
gcgatcggta aaattcgtgt tgaccacctg gagagcgaaa ttctgcgtgc gaaagcggag 900
ggtgcgggtc cgttcatggt tagcgcgacc gcgggtacca ccgtgattgg tgcgtttgac 960
ccgtggaac agattgcgga tctgtgcaag aaatacaacc tgtggatgca tgttgatgac 1020
gcgtgggggt gcgggtgcgt gatgagcaag aaatatcgta gcctgctgaa aggtatcgaa 1080
cgtagcgaca gcgttacctg gaaccgcac aagctgctgg cggcgccgca gcaatgcagc 1140
accttctga cccgtcacga gggcattctg agcgaatgcc acagaccaa cgcgacctac 1200
ctgttcaga aggacaaatt ttacgatacc caatatgaca ccggtgataa acacatccag 1260
tcgggccgtc gtgcggacgt gctgaaattc tggtttatgt ggctgcgaa gggtagcagc 1320
ggtctggagc aacacatcga taaagtttcc gagaacgcgg aacactttac caacagcatt 1380
aaggcgcgtg acggtttcga aatggtggtt gagaccccggt aatgcaccaa cgtgtgcttt 1440
tggtatgttc cgcgggtctc gcgtagcgtg ccgcgtgata gcgcggagtt caccgaacgt 1500

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ctgcacaagg tggcgccgaa ggttaagag cgtatgatgc gtgaaggtag catgatgatc	1560
acctaccagc cgattcacga caaacccaac ttctttcgtc tggttctgca aaacagcgcg	1620
ctggacaaga gcgatatgaa ctatatcatt gatgagatcg aacgtctggc ggcggaacctg	1680
taa	1683

<210> SEQ ID NO 26
 <211> LENGTH: 1689
 <212> TYPE: DNA
 <213> ORGANISM: Aedes aegypti

<400> SEQUENCE: 26

atgccggcga acggcatggt cgtatgtggc ctgcaagtta tcgacgatag caacgtgagc	60
agcggtagcg acagcgcggg cgtgagcgag gatgaagatg ttcaactggt ttgctcgaag	120
ggcaacacca ttgttccgaa accgctgaag aaaagcatca gcaagattaa agatgaggaa	180
tttagcaaga ccgcgaaagc gaacgagaaa cgttacgcga gcctgccgag ccgtgaacac	240
caccagcaat tctgaccga ctttctgagc gaggtgctga acaacgcggt tttcaacgcg	300
accgaacgtg cgaacaagggt gctgaactgg gttgatccgg agcagctgaa gcgtacctg	360
gacctggagc tgaagacga accggatagc cagcagaagc tgctggaact gacctgtgcg	420
accatcaagc acagcgtgaa aaccggtcac ccgtacttca tgaaccagct gtttagcagc	480
gttgatccgt atggttttgc gggccaaatt ctgaccgacg cgctgaaccc gagcgtgtac	540
accttcgaag ttgccccggt gtttgttctg atggagggaag tggttctgaa agaatgcgt	600
accatcgtgg gttaccgga ccgacccggt gatggcattt tctgcccggt tggcagcatg	660
gcgaacggtt atagcatcag ctgcgcgcgt tttaagcaca tgccggatgt taagacccaa	720
ggcctgcaca gcctgccggt tctggtgatt ttcaccagcg aagacgcgca ctacagcgtt	780
aagaaactgg cgagctttat gggtatcggc agcgacacg tgatatccgat tcgtaccgat	840
gcgatcggta aaattcgtgt tgaccacctg gagagcgaaa ttctgcgtgc gaagagcgag	900
ggtgcggtgc cgttcattgt tagcgcgacc gcgggtacca ccgtgattgg tgcgtttgac	960
ccgctggaac agattgcgga tctgtgcaag aaatacaacc tgtggatgca tgttgatgcg	1020
gcgtggggtg gcggtgcgct gatgagcaag aaatatcgta gcctgctgaa aggtatcgag	1080
cgtagcgaca gcgtgacctg gaaccgcgac aagctgctgg cgccgcgca gcaatgcagc	1140
accttctcga ccggtcacga gggcattctg agcgaatgcc acagcaccaa cgcgacctac	1200
ctgttcaga aggacaaatt ttacgatacc caatatgaca ccggtgataa acacatccag	1260
tcgcgccgtc gtgcggatgt gctgaaattc tggtttatgt ggctgctgaa gggtagcagc	1320
ggcctggaac aacacatcga caaagtttct gagaacgcgg aacactttac cagcagcatt	1380
aaggcgcgtg agggtttcga aatggtggtt gagaacccgg aatgcaccaa cgtgtgcttt	1440
tggtatgttc cgccgggtct gcgtaactg ccgctgata gcgcggagtt caccgaacgt	1500
ctgcacaaag tggcgccgaa ggttaagag cgtatgatgc gtgaaggtag catgatgatc	1560
acctaccagc cgattcacga caaacccaac ttctttcgtc tggttctgca aaacagcgcg	1620
ctggacaaga gcgatatgaa ctatatcatt gatgagatcg aacgtctggc ggcggaacctg	1680
aagccgtaa	1689

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<210> SEQ ID NO 27
 <211> LENGTH: 1719
 <212> TYPE: DNA
 <213> ORGANISM: *Anopheles sinensis*

<400> SEQUENCE: 27

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atgccggcga acggtgtgaa cagcgttgag ctggaagtga tgcaggatgt tgcgaccacc      60
tacgcgagcg gtagcgacag cgcgggcgtg agcgaggatg aagatgtgca gcaactgttc      120
gttagcgggtg cgcaccacat tagcagcgtg ccgccgctga agaaagcggg tgagaccctg      180
ggcaagggtta cccagctgca aggtccggcg agcgagggtg cggcggcggc ggaagtgagc      240
gaaaaacgtt atgcgagcct gccgaaccgt gagcagcacc agcaattcct gaccgatatt      300
ctgaccgaag tgctgaacag cgcgggttttc aacgcgaccg atcgtgcgaa caaggtgctg      360
aactgggttg acccggagga actgaagcgt accctggacc tggcgatcaa acaagagccg      420
gatacccacg agaagctgct ggaactgacc cgtgcgacca ttcgtcacag cgtgaaaacc      480
ggtcaccctg acttcatgaa ccagctgttc agcagcgtgg acccgtagcg ttttgcgggc      540
caagtcttga cgcagcgcgt gaacccgagc gtgtacacct tcgaagttag cccggtgttt      600
gttctgatgg aggaagtggg tctgcgtgag atgcgtacca tcgtgggtta tccgaacggc      660
gaaggtgacg gcatttttcg gccgggtggc agcatggcga acggttacgc gatcagctgc      720
gcgcgttata agtttatgcc ggatgttaaa gcgaaaggtc tgcatgcgct gccgcgtctg      780
gtgattttca ccagcgaaga cgcgcactac agcgttaaga aactggcgag ctttatgggt      840
atcggcagcg acaacgtgta tgcgattaag accgatgcga tcggtaaaaa ttgcgttgac      900
cacctggaga gcgaaatcct gcgtgcgaag caggaaagggt gcgtgccgtt catggtttag      960
gcgaccgcgg gtaccaccgt tattggtgcg ttgacccgc tggaacaaat tgcggatctg     1020
tgcgcgaaat acaacctgtg gatgcattgt gatgcggcgt ggggtggcgg tgcgctgatg     1080
agcaagaaat atcgtacctt gctgaaagggt atcgagcgtg gcgacagcgt tacctggaac     1140
ccgcacaagc tgctggcggc gccgcagcaa tgcagacccc tgctgacccg tcaccgtaac     1200
attctgagcg aatgccacag caccaacgcg acctacctgt tccagaagga caaatattac     1260
gatacccgtt atgacaccgg tgataaacac atccaatgcg gccgtcgtgc ggatgttctg     1320
aaattcttgt ttatgtggcg tgcgaagggt accgcgggct ttgagcagca cattgacaaa     1380
gtgttcgaga acgcggaaca ctttaccagc agcatcaagg cgcgtccggg tttcgaaatg     1440
gttattgaga acccggaatg caccaacgtg tgcttttggt atgttccgcc gggctctgcgt     1500
agcgtgccgc gtgatagcgc ggagtttctg gaacgtctgc acaaagtggc gccgaagggt     1560
aaagagcgta tgatgaagga aggttagcatg atgatcacct accagccgat tcacgacaaa     1620
ccgaacttct ttcgtctggt tctgcaaaac agcagcctgg acaagagcga tatgaactat     1680
atcattgatg agatcgaacg tctgggcaaa gacctgtaa                                1719

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<210> SEQ ID NO 28
 <211> LENGTH: 15
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Conserved N-Term 15-aa mosquito ADC sequence

<400> SEQUENCE: 28

Ser Gly Ser Asp Ser Ala Gly Val Ser Glu Asp Glu Asp Val Gln

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1	5	10	15
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<210> SEQ ID NO 29
<211> LENGTH: 491
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
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<223> OTHER INFORMATION: Consensus Sequence
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<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (4)..(4)
<223> OTHER INFORMATION: X is N, S, I, V, or D
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (5)..(5)
<223> OTHER INFORMATION: X is R or Q
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (6)..(6)
<223> OTHER INFORMATION: X is E, S, A, R, K, or absent
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (7)..(7)
<223> OTHER INFORMATION: X is Q, H, V, S, M, L, or R
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (9)..(9)
<223> OTHER INFORMATION: X is Q, E, or F
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (10)..(10)
<223> OTHER INFORMATION: X is R, Q, T, K, or E
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (12)..(12)
<223> OTHER INFORMATION: X is L or I
<220> FEATURE:
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<222> LOCATION: (13)..(13)
<223> OTHER INFORMATION: X is T, R, or K
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (14)..(14)
<223> OTHER INFORMATION: X is D, E, Q, N, K, or S
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (15)..(15)
<223> OTHER INFORMATION: X is F, C, A, M, L, S, or A
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (16)..(16)
<223> OTHER INFORMATION: X is L, I, F, V, or M
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (17)..(17)
<223> OTHER INFORMATION: X is S, T, E, or D
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (18)..(18)
<223> OTHER INFORMATION: X is E, V, N, S, I, or L
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (19)..(19)
<223> OTHER INFORMATION: X is V, I, L, or M
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (21)..(21)
<223> OTHER INFORMATION: X is N, S, K, Q, or E
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (22)..(22)
<223> OTHER INFORMATION: X is G, N, S, A, K, E, or D
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (24)..(24)
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<223> OTHER INFORMATION: X is I or V
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (26)..(26)
<223> OTHER INFORMATION: X is N, S, E, D, or Q
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (27)..(27)
<223> OTHER INFORMATION: X is A, G, or D
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (29)..(29)
<223> OTHER INFORMATION: X is D, E, K, R, N, or S
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (31)..(31)
<223> OTHER INFORMATION: X is S, A, K, E, or N
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (33)..(33)
<223> OTHER INFORMATION: X is K, R, P, or L
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (35)..(35)
<223> OTHER INFORMATION: X is L or V
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (36)..(36)
<223> OTHER INFORMATION: X is N, E, D, or Q
<220> FEATURE:
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<222> LOCATION: (37)..(37)
<223> OTHER INFORMATION: X is W or F
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (38)..(38)
<223> OTHER INFORMATION: X is V, K, R, H, or T
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
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<223> OTHER INFORMATION: X is D, N, Q, E, C, or A
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (41)..(41)
<223> OTHER INFORMATION: X is E, K, or N
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (42)..(42)
<223> OTHER INFORMATION: X is E, Q, I, or V
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (44)..(44)
<223> OTHER INFORMATION: X is K, Q, S, E, A, or R
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (45)..(45)
<223> OTHER INFORMATION: X is R, N, E, K, or S
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (46)..(46)
<223> OTHER INFORMATION: X is S, T, I, R, L, or K
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (47)..(47)
<223> OTHER INFORMATION: X is I, L, F, V, or M
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (48)..(48)
<223> OTHER INFORMATION: X is D, N, or E
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (49)..(49)
<223> OTHER INFORMATION: X is L, F, or M
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (50)..(50)

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<223> OTHER INFORMATION: X is S, E, A, K, G, or D
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (51)..(51)
<223> OTHER INFORMATION: X is L, I, or V
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (52)..(52)
<223> OTHER INFORMATION: X is K, R, or S
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (53)..(53)
<223> OTHER INFORMATION: X is D, A, Q, S, T, G, or N
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
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<223> OTHER INFORMATION: X is E, N, T, D, A, or G
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (55)..(55)
<223> OTHER INFORMATION: X is P, S, A, or R
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (56)..(56)
<223> OTHER INFORMATION: X is D, S, or V
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (57)..(57)
<223> OTHER INFORMATION: X is S or T
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (58)..(58)
<223> OTHER INFORMATION: X is D or H
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (59)..(59)
<223> OTHER INFORMATION: X is E, Q, or D
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (60)..(60)
<223> OTHER INFORMATION: X is K, R, E, T, or D
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (62)..(62)
<223> OTHER INFORMATION: X is L or M
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (63)..(63)
<223> OTHER INFORMATION: X is E, Q, L, or K
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (64)..(64)
<223> OTHER INFORMATION: X is L, T, V, or I
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (65)..(65)
<223> OTHER INFORMATION: X is A, T, L, I, or V
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (66)..(66)
<223> OTHER INFORMATION: X is R, K, L, D, or N
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (67)..(67)
<223> OTHER INFORMATION: X is A, D, E, or K
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (68)..(68)
<223> OTHER INFORMATION: X is T, M, V, or I
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (69)..(69)
<223> OTHER INFORMATION: X is I or V
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (70)..(70)

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<223> OTHER INFORMATION: X is D, K, R, T, N, or H
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (71)..(71)
<223> OTHER INFORMATION: X is H or Y
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (79)..(79)
<223> OTHER INFORMATION: X is Y or F
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (81)..(81)
<223> OTHER INFORMATION: X is M or V
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (87)..(87)
<223> OTHER INFORMATION: X is S, G, C, or A
<220> FEATURE:
<221> NAME/KEY: MISC_FEATURE
<222> LOCATION: (88)..(88)
<223> OTHER INFORMATION: X is V or L
<220> FEATURE:
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<223> OTHER INFORMATION: X is K, H, R, or N
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<223> OTHER INFORMATION: X is H, R, F, Y, V, or A
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<223> OTHER INFORMATION: X is V, I, A, or M
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<223> OTHER INFORMATION: X is I, M, or L
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<223> OTHER INFORMATION: X is A or S
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<223> OTHER INFORMATION: X is F or L
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<223> OTHER INFORMATION: X is I or L
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<223> OTHER INFORMATION: X is D, C, or N
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<223> OTHER INFORMATION: X is V, L, S, I, or A

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<223> OTHER INFORMATION: X is S or V

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<223> OTHER INFORMATION: X is A or S

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<223> OTHER INFORMATION: X is K or R
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<223> OTHER INFORMATION: X is A, S, Q, or N
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<223> OTHER INFORMATION: X is R, Q, or T
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<223> OTHER INFORMATION: X is G or D
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<223> OTHER INFORMATION: X is E, V, or R
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<223> OTHER INFORMATION: X is M or L
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<223> OTHER INFORMATION: X is N, T, Q, H, E, K, or P
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Xaa Val Xaa Xaa Xaa Xaa Xaa Pro Xaa Xaa Leu Xaa Xaa Xaa Xaa Xaa
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Xaa Xaa Xaa Xaa Xaa Xaa Xaa Xaa Xaa Xaa Xaa Xaa Leu Xaa Xaa Xaa
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Xaa Xaa Xaa Xaa Xaa Xaa Xaa Ser Val Lys Thr Gly His Pro Xaa Phe
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Xaa Asn Gln Leu Phe Ser Xaa Xaa Asp Xaa Tyr Xaa Xaa Xaa Xaa Gln
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Xaa Xaa Xaa Asp Ala Leu Asn Pro Xaa Val Tyr Thr Xaa Glu Val Ser
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Pro Xaa Phe Xaa Leu Met Glu Glu Xaa Val Leu Xaa Glu Met Arg Xaa
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Gly Gly Ser Xaa Xaa Asn Gly Tyr Xaa Ile Ser Cys Xaa Arg Xaa Xaa
145 150 155 160

Xaa Xaa Pro Xaa Xaa Lys Xaa Lys Gly Leu Xaa Xaa Xaa Pro Arg Leu
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Val Xaa Phe Thr Ser Xaa Asp Xaa His Tyr Ser Xaa Xaa Lys Xaa Xaa
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Ser Xaa Xaa Gly Xaa Gly Xaa Asp Asn Val Tyr Xaa Xaa Xaa Xaa Xaa
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Xaa Xaa Gly Xaa Xaa Xaa Xaa Xaa Xaa Leu Xaa Xaa Xaa Xaa Xaa Arg
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Gly Thr Thr Val Xaa Gly Ala Phe Asp Pro Xaa Xaa Xaa Xaa Ala Xaa
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Xaa	Xaa	Glu	Xaa	Xaa	Xaa	Xaa	Xaa	Leu	Xaa	Xaa				
			485					490						

1. A recombinant truncated insect aspartate 1-decarboxylase (ADC), the truncated insect ADC lacking a sufficient number of contiguous residues within the amino terminal region of a corresponding full length wild-type insect ADC such that the truncated ADC exhibits increased conversion of aspartate to beta-alanine as compared to the corresponding full length wild-type insect ADC.

2. The recombinant truncated insect ADC of claim 1, which is a truncated variant of a mosquito, fly, beetle, flea, roach, or termite ADC; or which is a truncated variant of a wild-type ADC enzyme, the wild-type ADC enzyme being defined by an amino acid sequence at least 90, 91, 92, 93, 94, 95, 96, 97, 98, or 99% identical overall to the amino acid sequence of SEQ ID NO: 29.

3. The recombinant truncated insect ADC of claim 1, which is a truncated variant of an insect ADC from the genus: *Culex*, *Anopheles*, *Drosophila*, *Aethina*, *Aedes*, *Tribolium*, *Anopheles*, *Tenebrio*, *Asbolus*, or *Cryptotermes*.

4. The recombinant truncated insect ADC of claim 1, which is a truncated variant of an insect ADC from the species: *Culex tarsalis*, *Anopheles arabiensis*, *Drosophila melanogaster*, *Culex quinquefasciatus*, *Aethina tumida*, *Aedes albopictus*, *Aedes aegypti*, *Tribolium castaneum*, *Anopheles sinensis*, *Tenebrio molitor*, *Asbolus verrucosus*, or *Cryptotermes secundus*.

5. The recombinant truncated insect ADC of claim 1, wherein the corresponding full length wild-type insect ADC is:

- (a) a mosquito ADC comprising an amino acid sequence at least 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, or 99% identical overall to any one of SEQ ID NOs: 2, 4, or 9-15;
- (b) a beetle ADC comprising an amino acid sequence at least 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, or 99% identical overall to any one of SEQ ID NOs: 1, 3, or 5-6;
- (c) a fly ADC comprising an amino acid sequence at least 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, or 99% identical overall to SEQ ID NO: 8; or
- (d) an ADC comprising an amino acid sequence at least 90, 91, 92, 93, 94, 95, 96, 97, 98, or 99% identical overall to SEQ ID NO: 29.

6. The recombinant truncated insect ADC of claim 1, wherein the truncated ADC comprises an amino acid sequence at least 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, or 99% identical overall to:

- (a) position 72 to 561 of the amino acid sequence of CtADC set forth in SEQ ID NO: 2;

- (b) position 79 to 568 of the amino acid sequence of AaADC set forth in SEQ ID NO: 4;
- (c) position 56 to 544 of the amino acid sequence of AtADC set forth in SEQ ID NO: 3;
- (d) position 52 to 540 of the amino acid sequence of TcADC set forth in SEQ ID NO: 1;
- (e) position 71 to 560 of the amino acid sequence of Aa2ADC set forth in SEQ ID NO: 10;
- (f) position 71 to 562 of the amino acid sequence of Aa3ADC set forth in SEQ ID NO: 11;
- (g) position 74 to 563 of the amino acid sequence of CqADC set forth in SEQ ID NO: 9;
- (h) position 72 to 561 of the amino acid sequence of Aa4ADC set forth in SEQ ID NO: 13;
- (i) position 74 to 624 of the amino acid sequence of AdADC set forth in SEQ ID NO: 14;
- (j) position 83 to 572 of the amino acid sequence of AsADC set forth in SEQ ID NO: 12;
- (k) position 72 to 561 of the amino acid sequence of As2ADC set forth in SEQ ID NO: 15;
- (l) position 53 to 541 of the amino acid sequence of TmADC set forth in SEQ ID NO: 6;
- (m) position 57 to 572 of the amino acid sequence of AvADC set forth in SEQ ID NO: 5; or
- (n) position 1 to 491 the amino acid sequence set forth in SEQ ID NO: 29.

7. The recombinant truncated insect ADC of claim 1, wherein the truncated ADC comprises a glycine residue at a position corresponding to position 96 of the amino acid sequence of CtADC set forth in SEQ ID NO: 2.

8. The recombinant truncated insect ADC of claim 1, wherein the truncated ADC lacks at least X contiguous residues of the amino terminus of the corresponding full length wild-type insect ADC, wherein X is any integer between 5 and 50.

9. The recombinant truncated insect ADC of claim 1, wherein the truncation occurs at a position immediately C-terminal (downstream) of a residue corresponding to position n of a full-length wild-type insect ADC, wherein n is any integer between 2 and Y, wherein Y is the most C-terminal residue position within the full-length wild-type insect ADC at which truncation from the N terminus can occur with the truncated ADC exhibiting increased conversion of aspartate to beta-alanine as compared to the full length wild-type ADC.

10. The recombinant truncated insect ADC of claim 1, wherein the truncation occurs at a position C-terminal (downstream) of a residue corresponding to any one of:

- (a) position 2, 3, 4, 5, 6, 7, 8, 9 10, or 11 to 71 of the amino acid sequence of CtADC set forth in SEQ ID NO: 2;
- (b) position 2, 3, 4, 5, 6, 7, 8, 9 10, or 11 to 78 of the amino acid sequence of AaADC set forth in SEQ ID NO: 4;
- (c) position 2, 3, 4, 5, 6, 7, 8, 9 10, or 11 to 55 of the amino acid sequence of AtADC set forth in SEQ ID NO: 3;
- (d) position 2, 3, 4, 5, 6, 7, 8, 9 10, or 11 to 51 of the amino acid sequence of TcADC set forth in SEQ ID NO: 1;
- (e) position 2, 3, 4, 5, 6, 7, 8, 9 10, or 11 to 70 of the amino acid sequence of Aa2ADC set forth in SEQ ID NO: 10;
- (f) position 2, 3, 4, 5, 6, 7, 8, 9 10, or 11 to 70 of the amino acid sequence of Aa3ADC set forth in SEQ ID NO: 11;
- (g) position 2, 3, 4, 5, 6, 7, 8, 9 10, or 11 to 73 of the amino acid sequence of CqADC set forth in SEQ ID NO: 9;
- (h) position 2, 3, 4, 5, 6, 7, 8, 9 10, or 11 to 73 of the amino acid sequence of Aa4ADC set forth in SEQ ID NO: 13;

- (i) position 2, 3, 4, 5, 6, 7, 8, 9 10, or 11 to 73 of the amino acid sequence of AdADC set forth in SEQ ID NO: 14;
- (j) position 2, 3, 4, 5, 6, 7, 8, 9 10, or 11 to 82 of the amino acid sequence of AsADC set forth in SEQ ID NO: 12;
- (k) position 2, 3, 4, 5, 6, 7, 8, 9 10, or 11 to 78 of the amino acid sequence of As2ADC set forth in SEQ ID NO: 15;
- (l) position 2, 3, 4, 5, 6, 7, 8, 9 10, or 11 to 52 of the amino acid sequence of TmADC set forth in SEQ ID NO: 6;
- (m) position 2, 3, 4, 5, 6, 7, 8, 9 10, or 11 to 56 of the amino acid sequence of AvADC set forth in SEQ ID NO: 5; or
- (n) position 1 to 491 the amino acid sequence set forth in SEQ ID NO: 29.

11. The recombinant truncated insect ADC of claim 1, wherein the truncation occurs at a position N-terminal (upstream) of a residue corresponding to any one of:

- (a) position 72 to 80 of the amino acid sequence of CtADC set forth in SEQ ID NO: 2;
- (b) position 79 to 87 of the amino acid sequence of AaADC set forth in SEQ ID NO: 4;
- (c) position 56 to 64 of the amino acid sequence of AtADC set forth in SEQ ID NO: 3;
- (d) position 52 to 60 of the amino acid sequence of TcADC set forth in SEQ ID NO: 1;
- (e) position 71 to 79 of the amino acid sequence of Aa2ADC set forth in SEQ ID NO: 10;
- (f) position 71 to 79 of the amino acid sequence of Aa3ADC set forth in SEQ ID NO: 11;
- (g) position 74 to 82 of the amino acid sequence of CqADC set forth in SEQ ID NO: 9;
- (h) position 72 to 82 of the amino acid sequence of Aa4ADC set forth in SEQ ID NO: 13;
- (i) position 74 to 82 of the amino acid sequence of AdADC set forth in SEQ ID NO: 14;
- (j) position 83 to 91 of the amino acid sequence of AsADC set forth in SEQ ID NO: 12;
- (k) position 72 to 87 of the amino acid sequence of As2ADC set forth in SEQ ID NO: 15;
- (l) position 53 to 61 of the amino acid sequence of TmADC set forth in SEQ ID NO: 6;
- (m) position 57 to 65 of the amino acid sequence of AvADC set forth in SEQ ID NO: 5; or
- (n) position 1 to 491 the amino acid sequence set forth in SEQ ID NO: 29.

12. The recombinant truncated insect ADC of claim 1, wherein the truncation occurs at a position N-terminal (upstream) of a residue corresponding to any one of:

- (a) position 75 of the amino acid sequence of CtADC set forth in SEQ ID NO: 2;
- (b) position 82 of the amino acid sequence of AaADC set forth in SEQ ID NO: 4;
- (c) position 59 of the amino acid sequence of AtADC set forth in SEQ ID NO: 3;
- (d) position 55 of the amino acid sequence of TcADC set forth in SEQ ID NO: 1;
- (e) position 74 of the amino acid sequence of Aa2ADC set forth in SEQ ID NO: 10;
- (f) position 74 of the amino acid sequence of Aa3ADC set forth in SEQ ID NO: 11;
- (g) position 77 of the amino acid sequence of CqADC set forth in SEQ ID NO: 9;
- (h) position 77 of the amino acid sequence of Aa4ADC set forth in SEQ ID NO: 13;

- (i) position 77 of the amino acid sequence of AdADC set forth in SEQ ID NO: 14;
- (j) position 86 of the amino acid sequence of AsADC set forth in SEQ ID NO: 12;
- (k) position 82 of the amino acid sequence of As2ADC set forth in SEQ ID NO: 15;
- (l) position 56 of the amino acid sequence of TmADC set forth in SEQ ID NO: 6;
- (m) position 60 of the amino acid sequence of AvADC set forth in SEQ ID NO: 5;
- (n) position 1 to 491 the amino acid sequence set forth in SEQ ID NO: 29.

13. A recombinant protein having aspartate 1-decarboxylase activity, the recombinant protein comprising an amino acid sequence at least 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, or 99% identical overall to:

- (a) position 72 to 561 of the amino acid sequence of CtADC set forth in SEQ ID NO: 2;
- (b) position 79 to 568 of the amino acid sequence of AaADC set forth in SEQ ID NO: 4;
- (c) position 56 to 544 of the amino acid sequence of AtADC set forth in SEQ ID NO: 3;
- (d) position 52 to 540 of the amino acid sequence of TcADC set forth in SEQ ID NO: 1;
- (e) position 71 to 560 of the amino acid sequence of Aa2ADC set forth in SEQ ID NO: 10;
- (f) position 71 to 562 of the amino acid sequence of Aa3ADC set forth in SEQ ID NO: 11;
- (g) position 74 to 563 of the amino acid sequence of CqADC set forth in SEQ ID NO: 9;
- (h) position 72 to 561 of the amino acid sequence of Aa4ADC set forth in SEQ ID NO: 13;
- (i) position 74 to 624 of the amino acid sequence of AdADC set forth in SEQ ID NO: 14;
- (j) position 83 to 572 of the amino acid sequence of AsADC set forth in SEQ ID NO: 12;
- (k) position 72 to 561 of the amino acid sequence of As2ADC set forth in SEQ ID NO: 15;

- (l) position 53 to 541 of the amino acid sequence of TmADC set forth in SEQ ID NO: 6;
- (m) position 57 to 572 of the amino acid sequence of AvADC set forth in SEQ ID NO: 5; or
- (n) position 1 to 491 the amino acid sequence set forth in SEQ ID NO: 29.

14. The recombinant protein of claim **13**, which comprises a glycine residue at a position corresponding to position 96 of the amino acid sequence of CtADC set forth in SEQ ID NO: 2.

15. A polynucleotide comprising a nucleic acid sequence encoding the recombinant truncated insect ADC as defined in claim **1**.

16. An expression cassette comprising the isolated or recombinant polynucleotide of claim **15** operably linked to a promoter that is heterologous with respect to the insect ADC.

17. A host cell that expresses the recombinant truncated insect ADC as defined in claim **1**.

18. The host cell of claim **17**, which is a bacterial, insect, mammalian, yeast or fungal cell.

19. (canceled)

20. A process for the production of beta-alanine, the process comprising:

- (a) providing an ADC enzyme source which is the truncated insect ADC as defined in claim **1**;
- (b) contacting the ADC enzyme source with a source of aspartate under conditions enabling the ADC enzyme source to catalyze the conversion of the aspartate to beta-alanine; and
- (c) isolating and/or concentrating the beta-alanine produced.

21. The process of claim **20**, wherein the ADC enzyme source is an intact host cell.

22. (canceled)

23. (canceled)

* * * * *